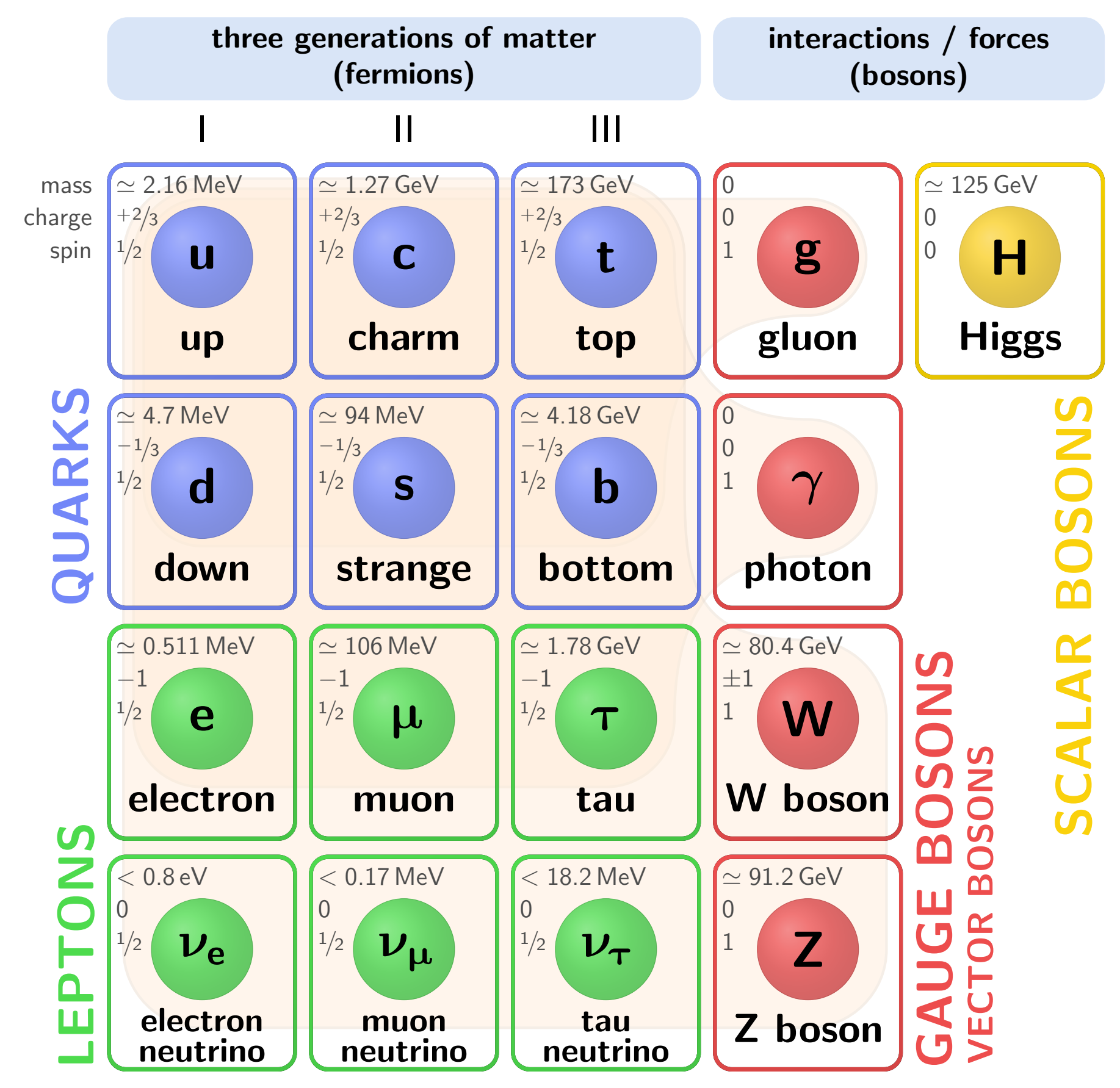
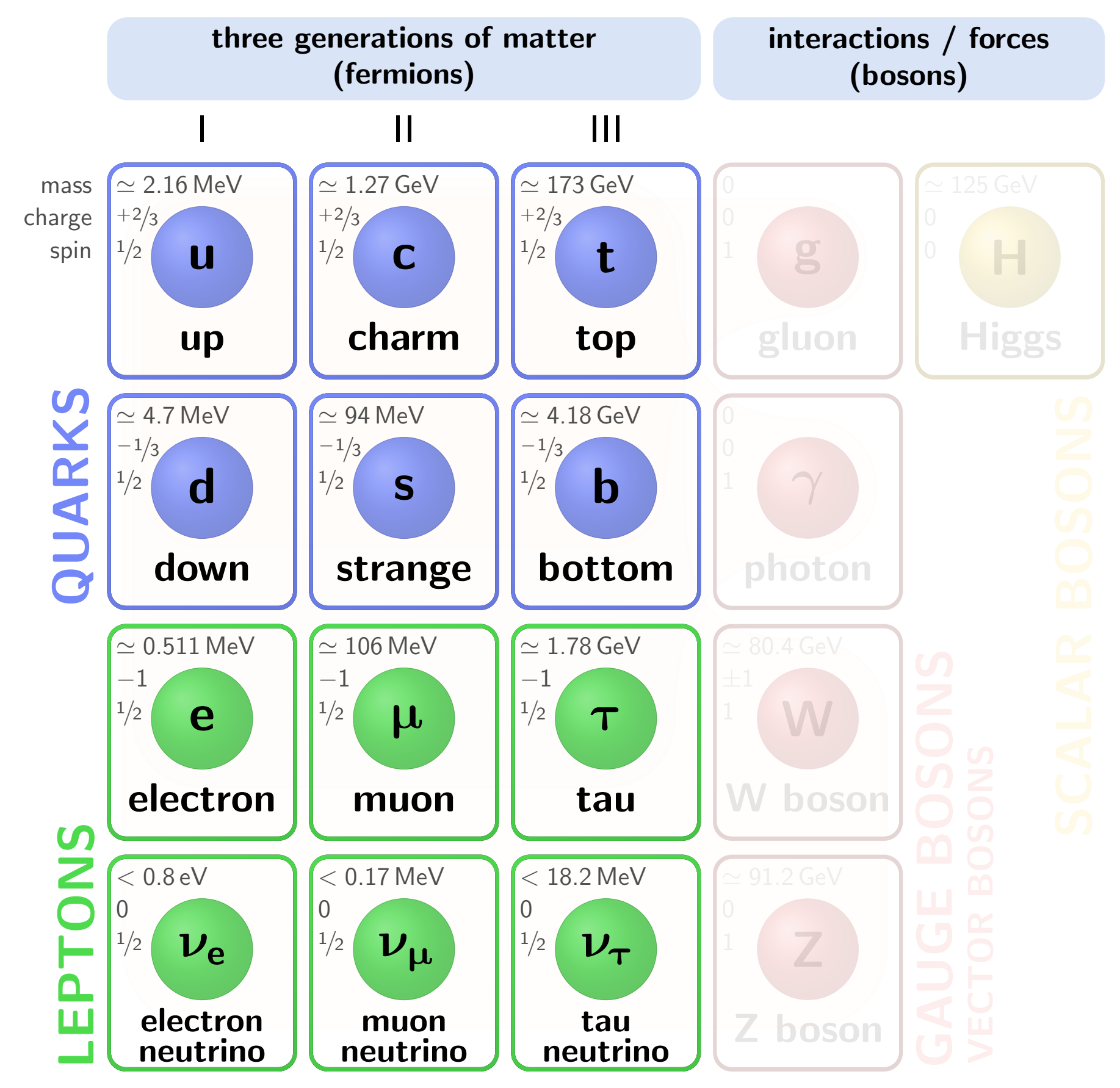
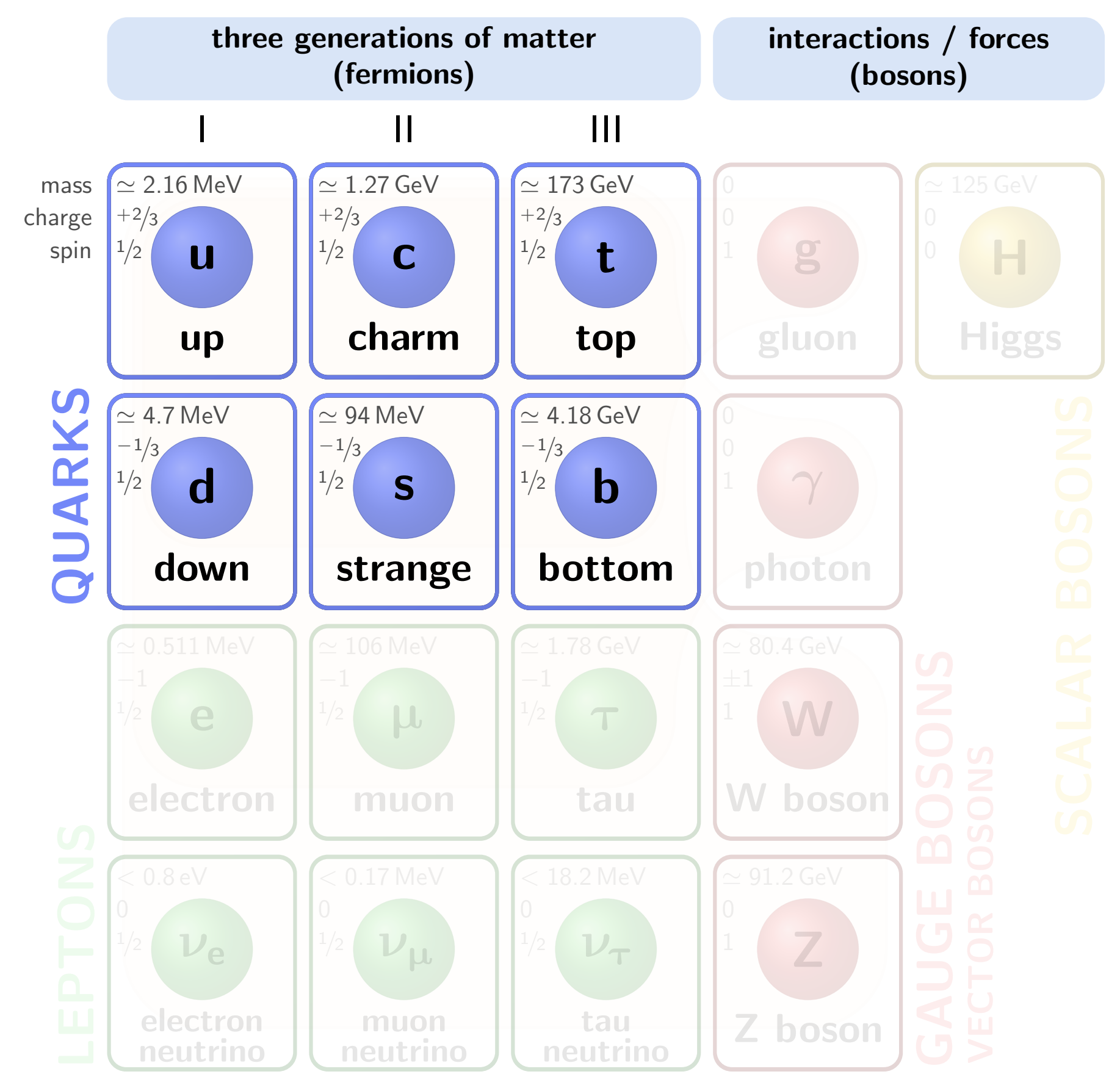
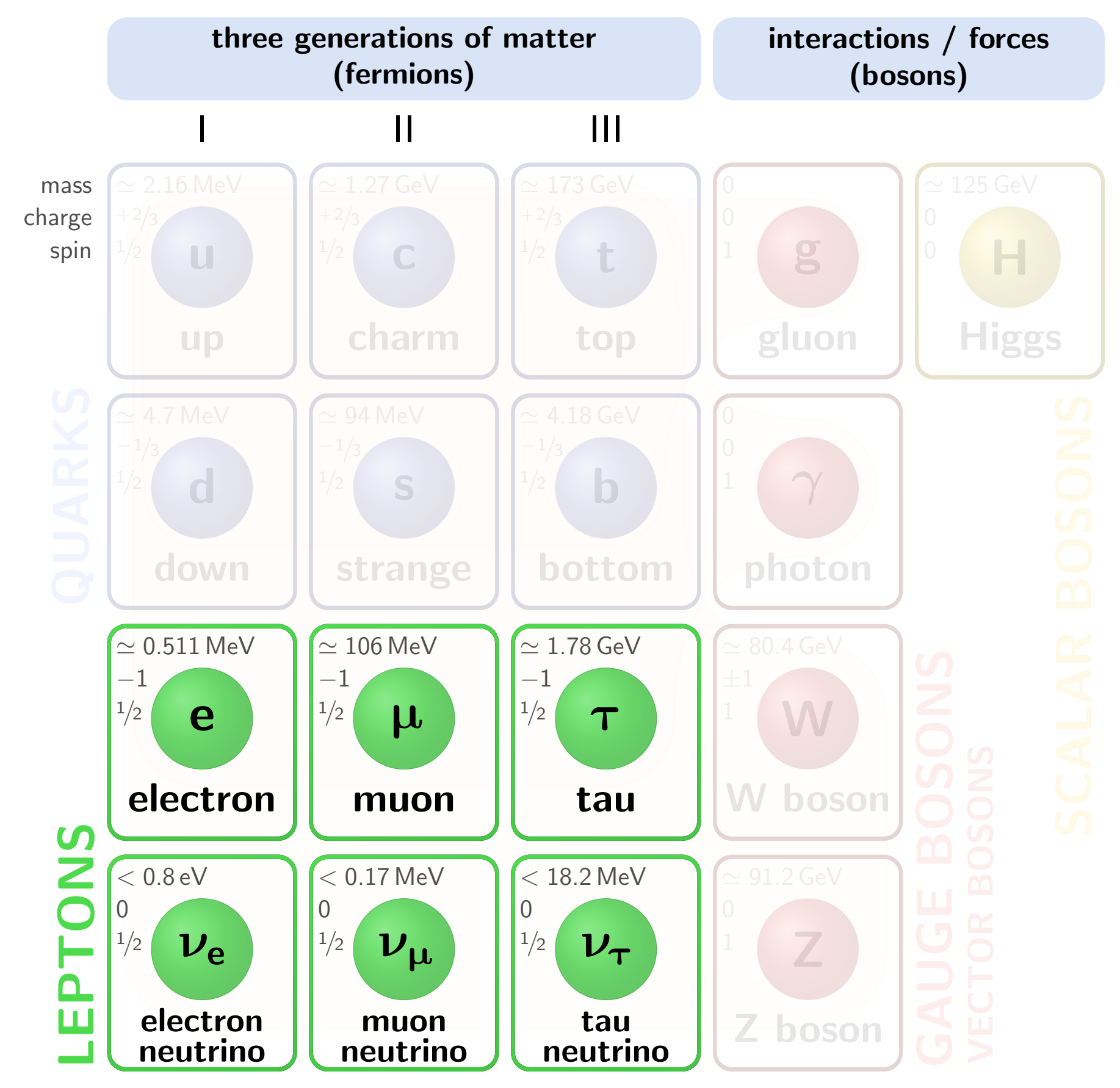
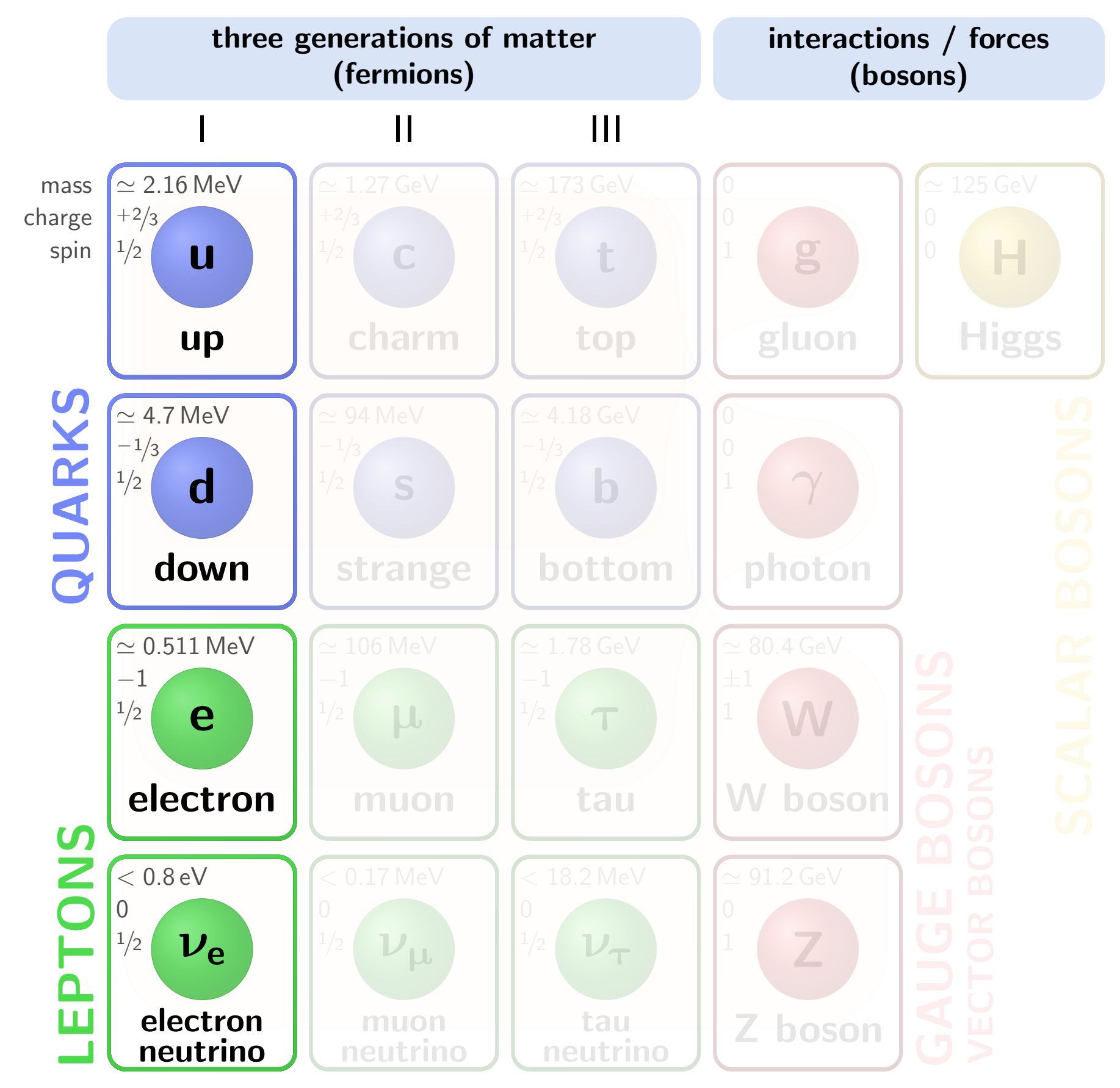
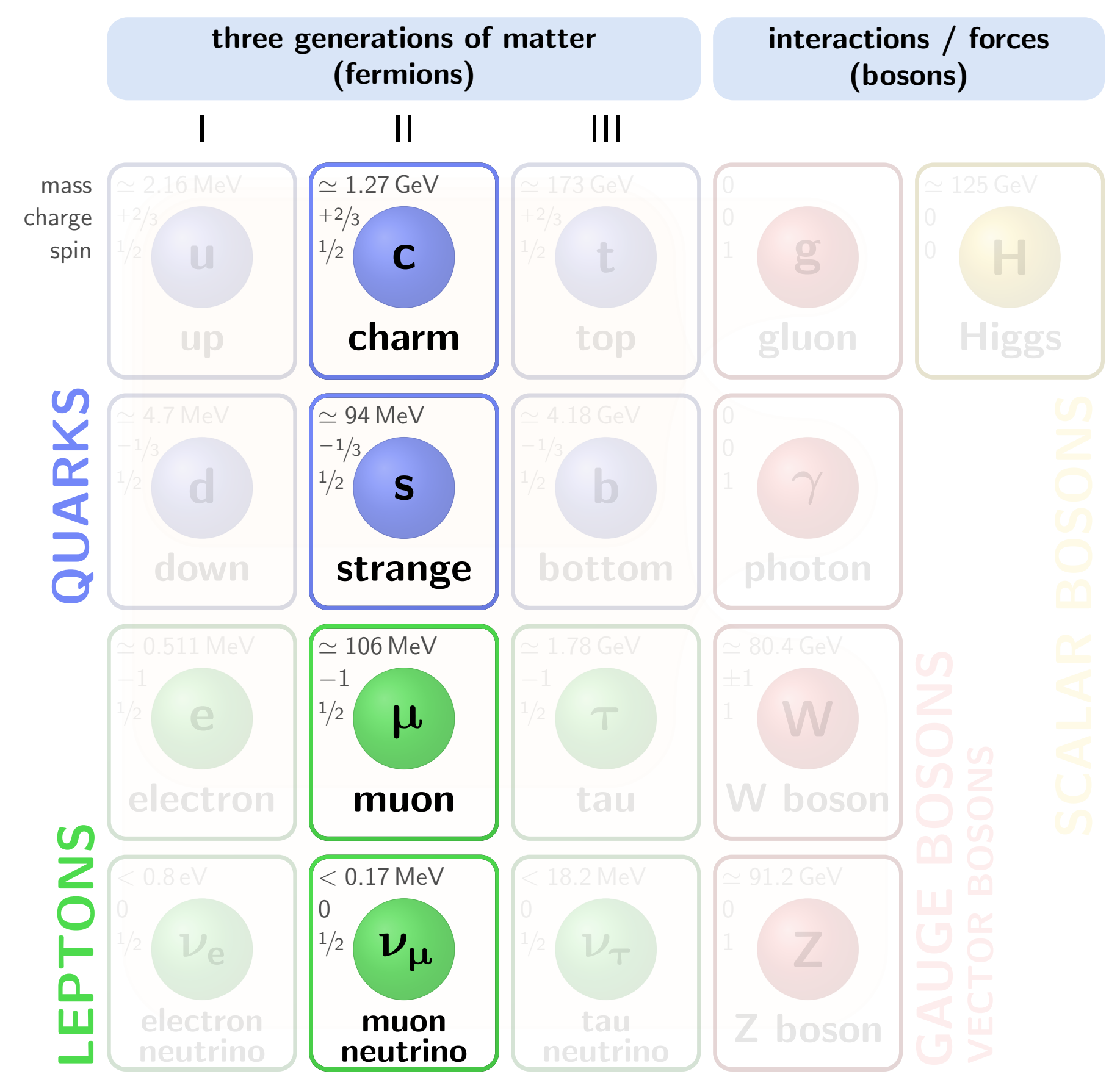

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
0
0
1
g
gluon
≃ 125 GeV
0
0
H
Higgs
QUARKS
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
−1/3
1/2
b
bottom
0
0
1
γ
photon
SCALAR BOSONS
≃ 0.511 MeV
−1
1/2
e
electron
≃ 106 MeV
−1
1/2
μ
muon
≃ 1.78 GeV
−1
1/2
τ
tau
≃ 80.4 GeV
±1
1
W
W boson
GAUGE BOSONS
VECTOR BOSONS
LEPTONS
< 0.8 eV
0
1/2
νe
electron
neutrino
< 0.17 MeV
0
1/2
νμ
muon
neutrino
< 18.2 MeV
0
1/2
ντ
tau
neutrino
≃ 91.2 GeV
0
1
Z
Z boson

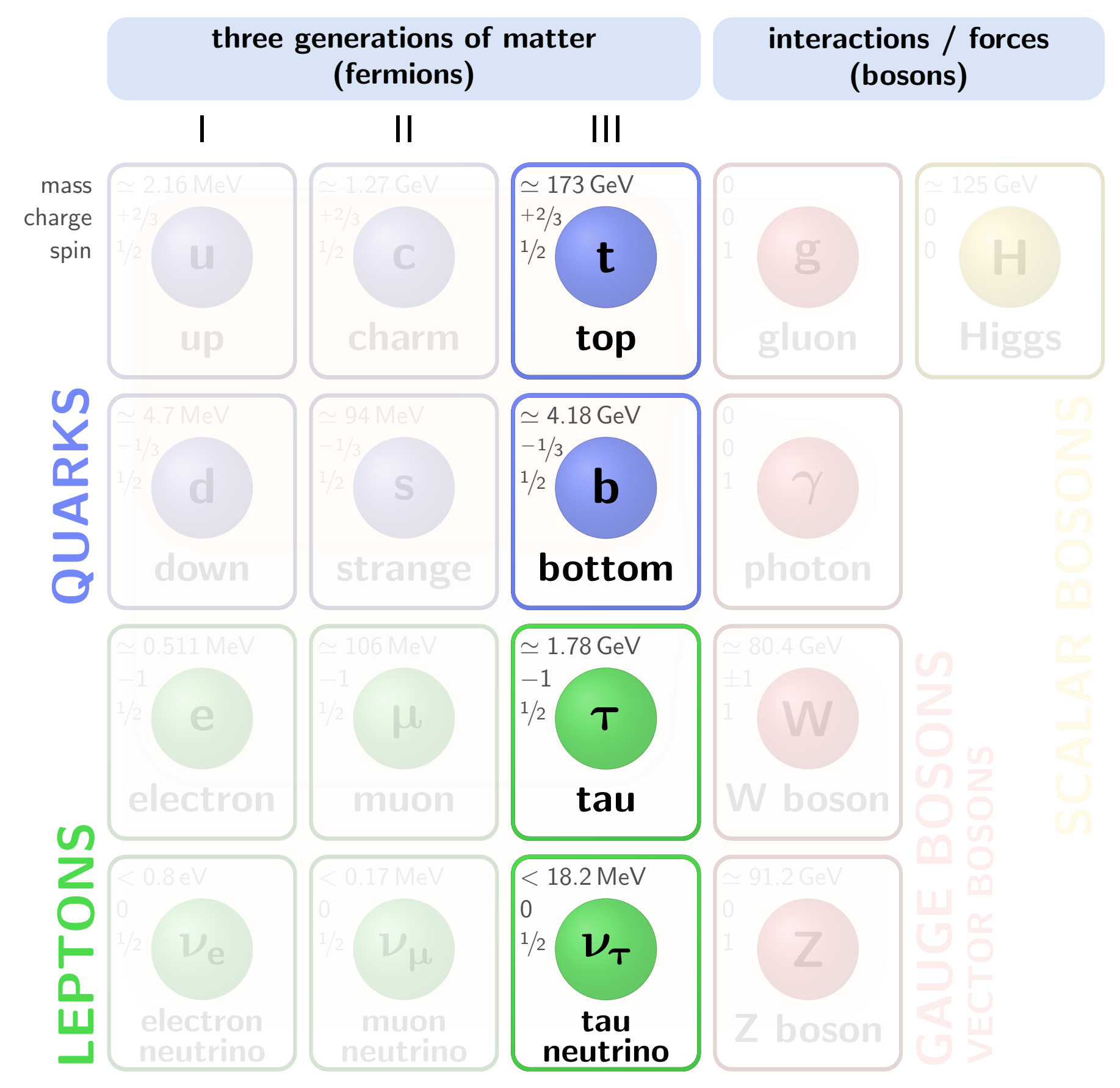

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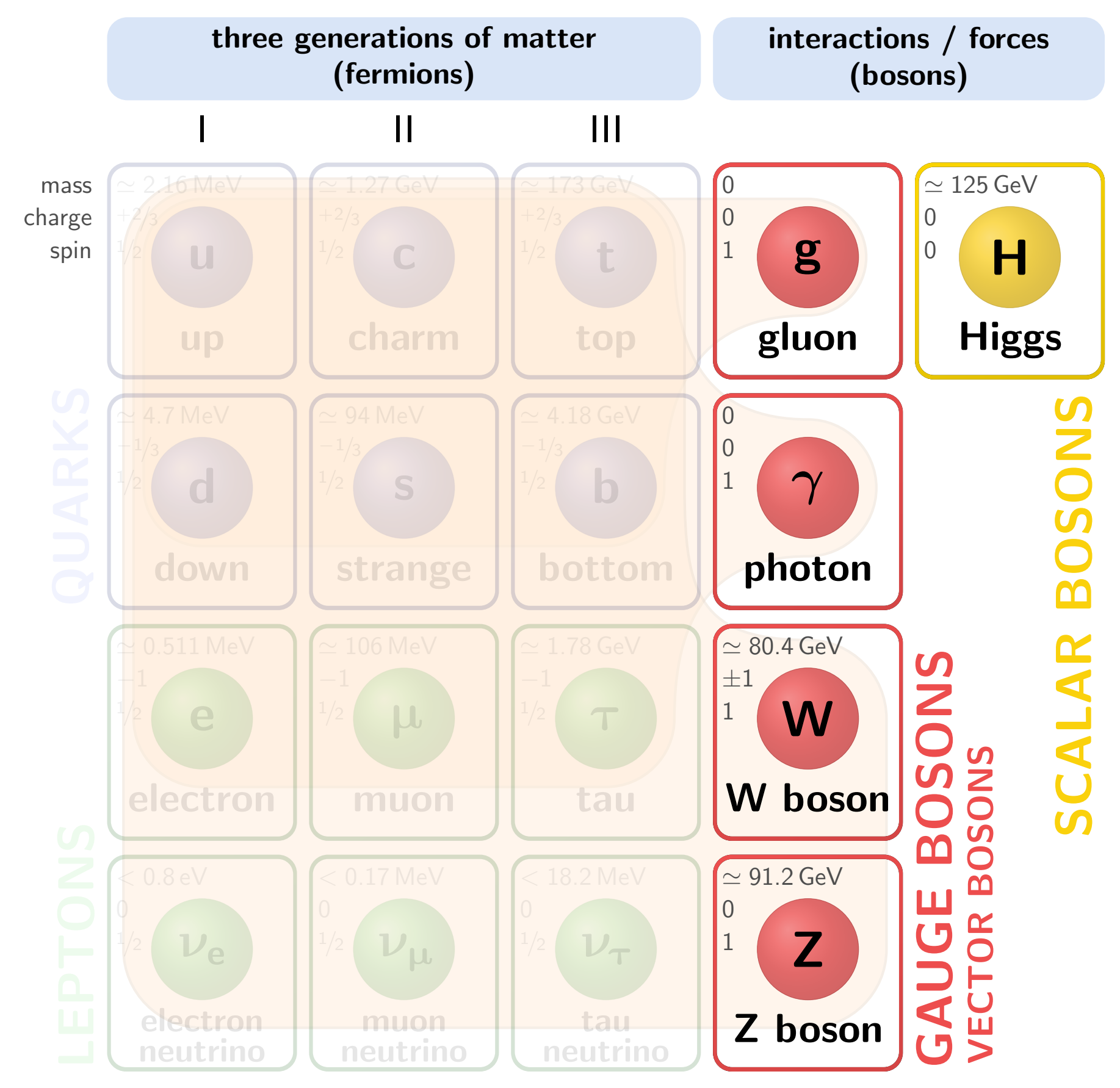

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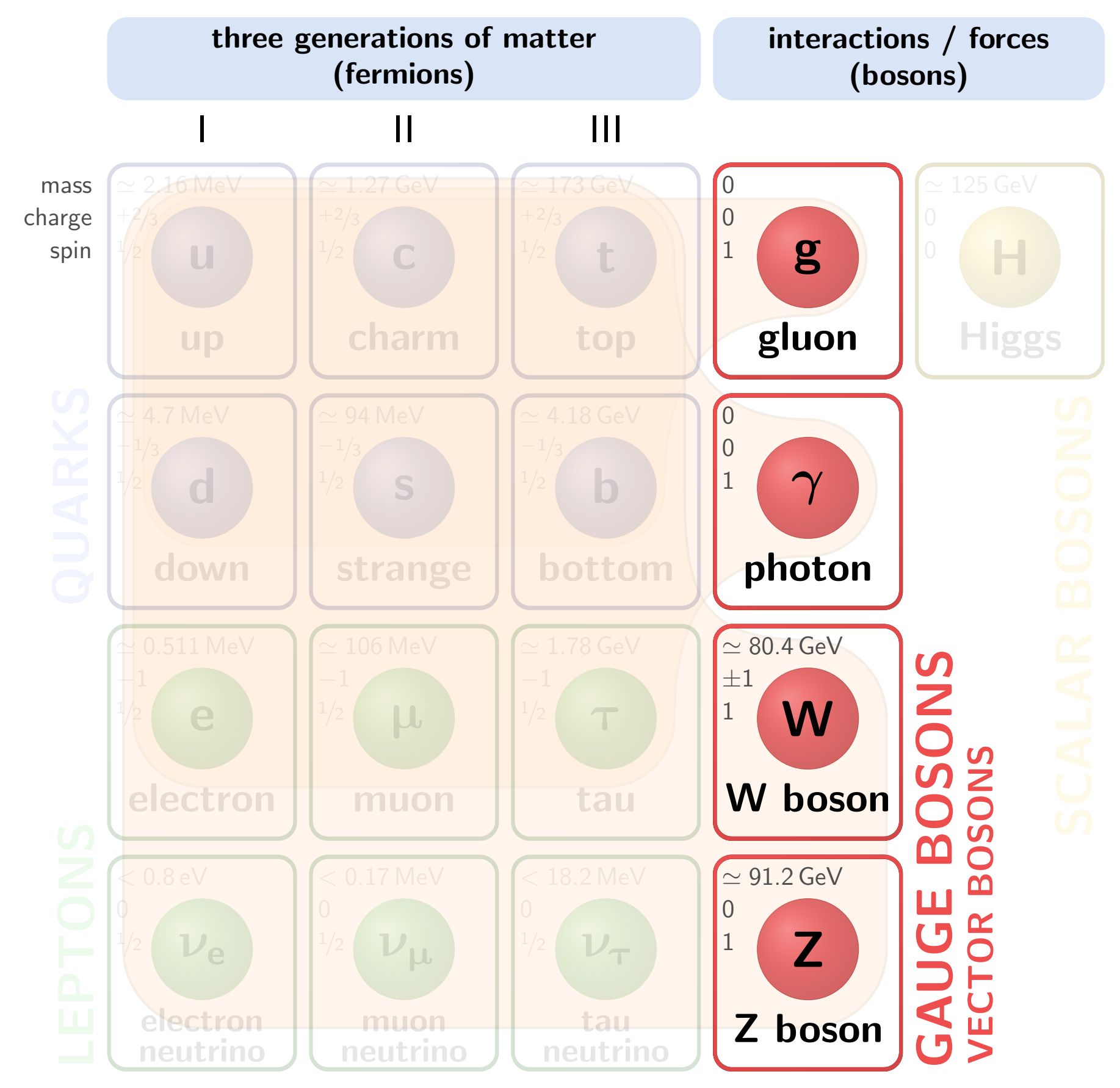

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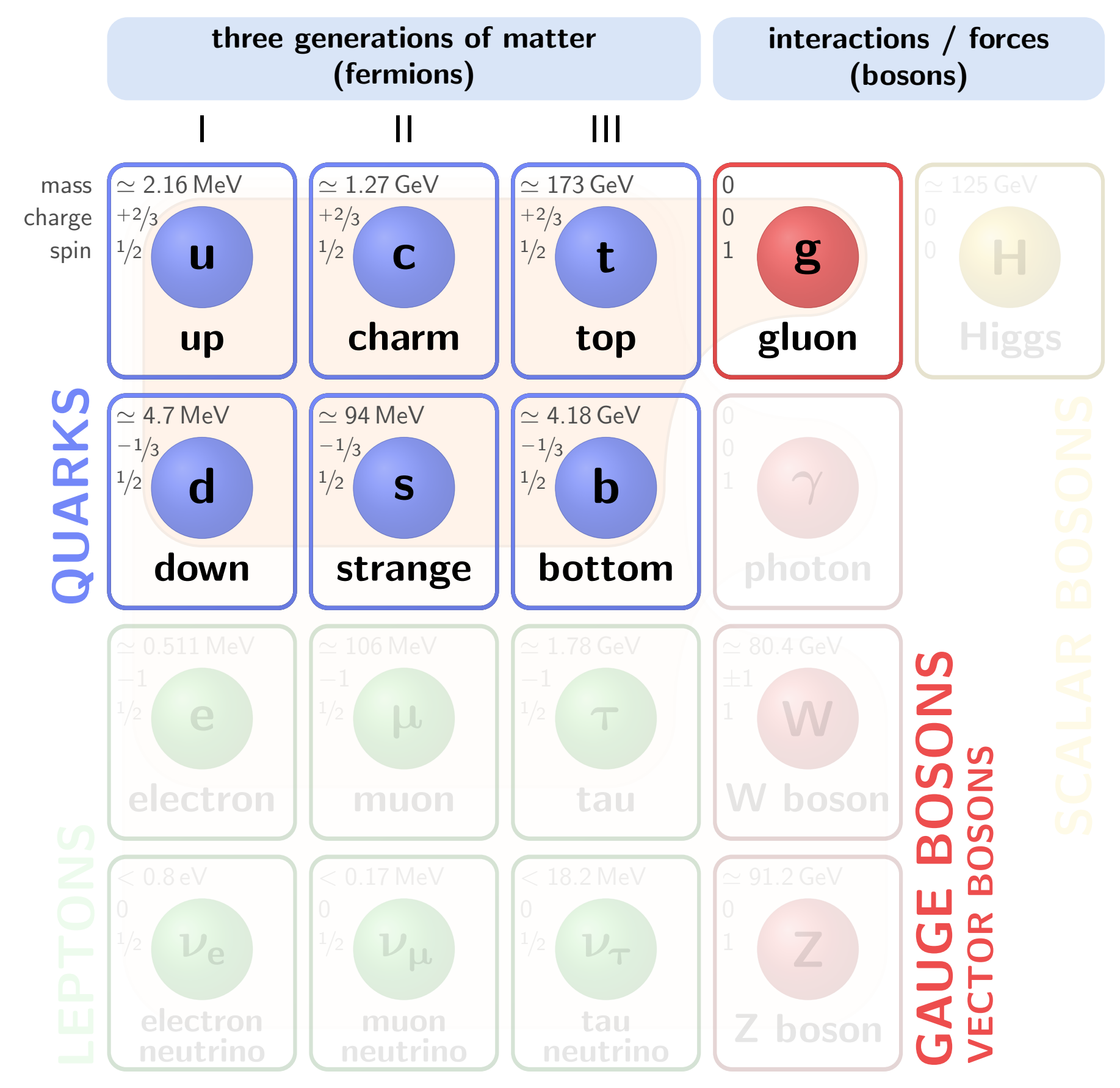

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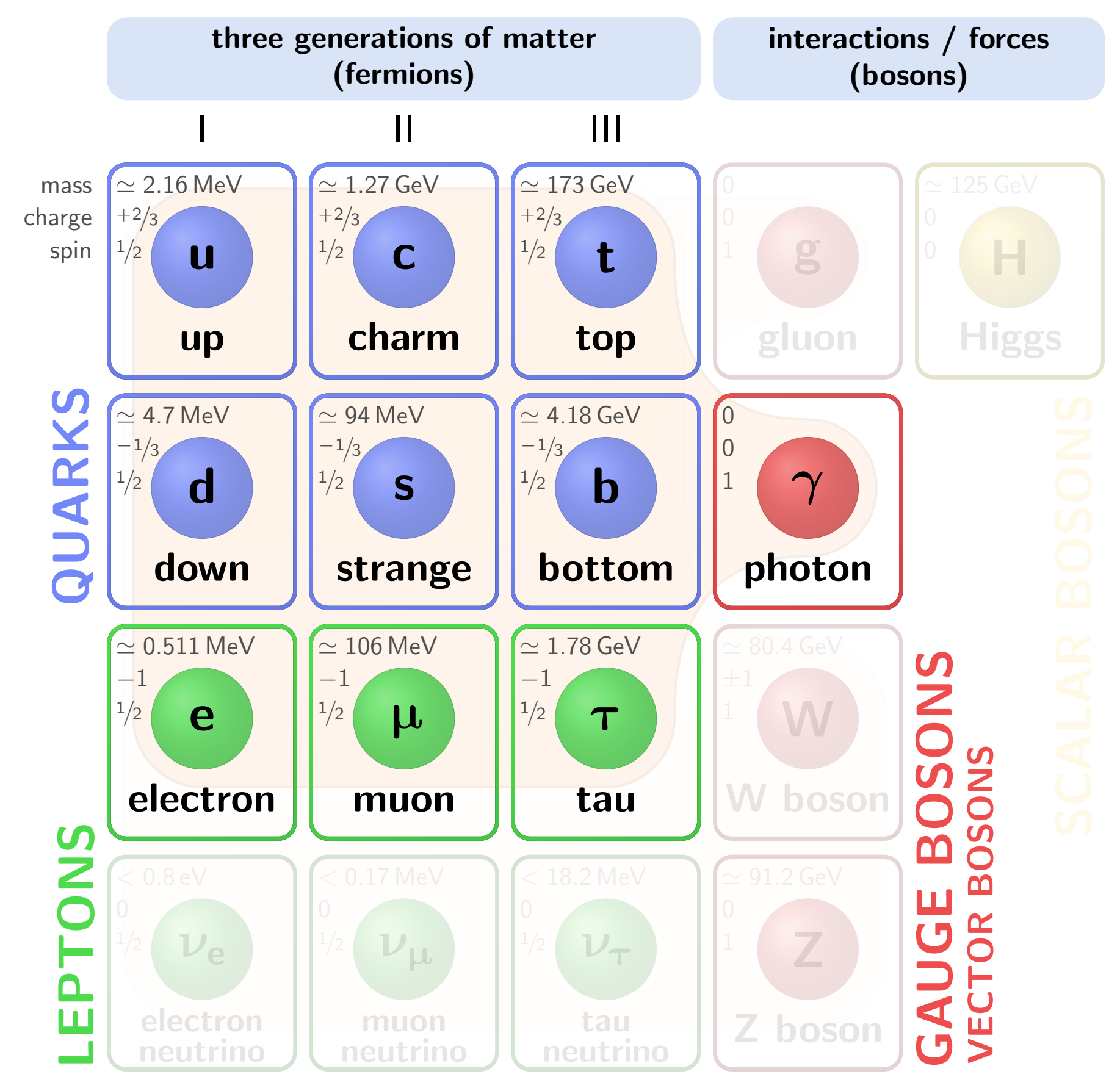

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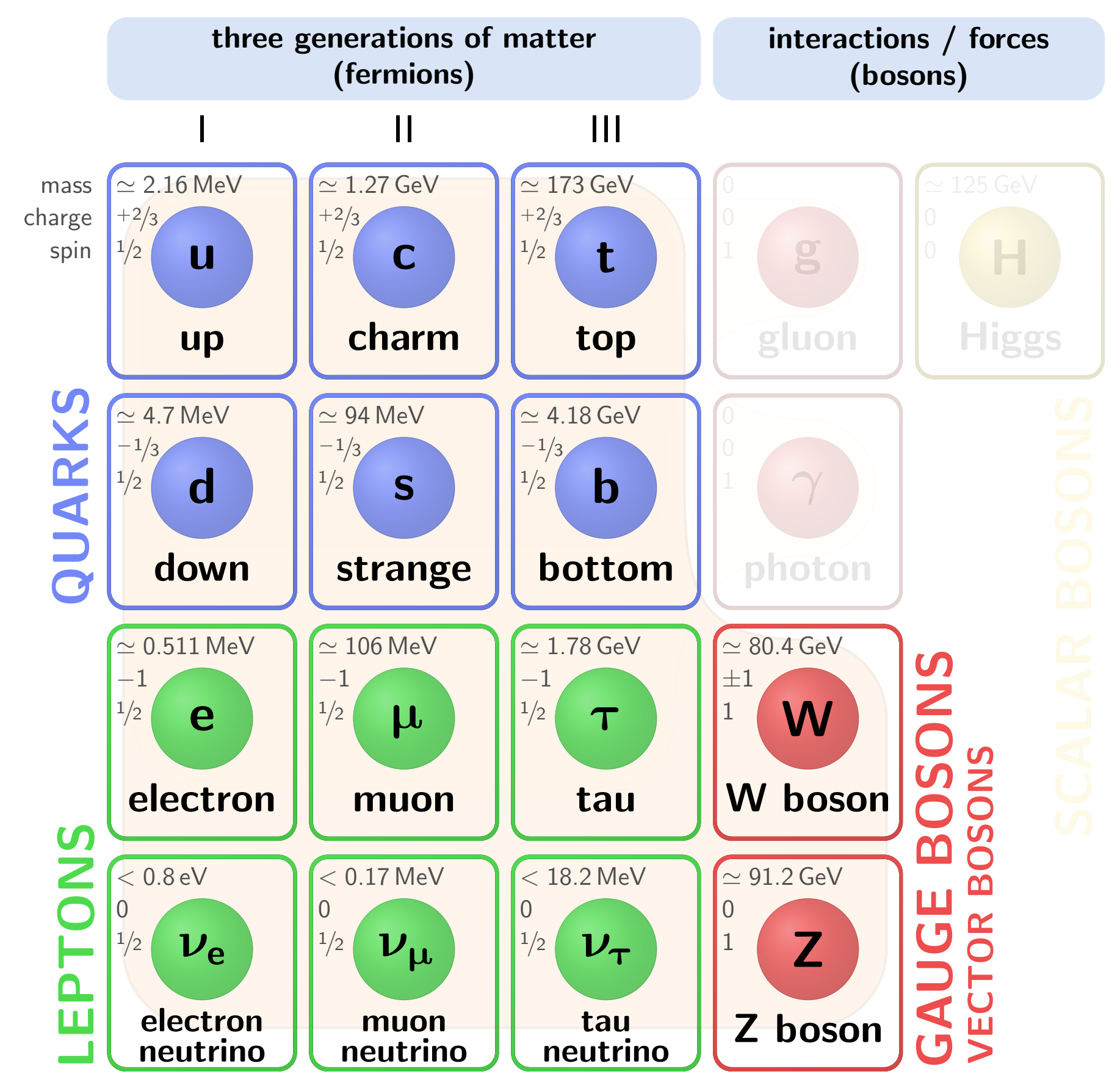

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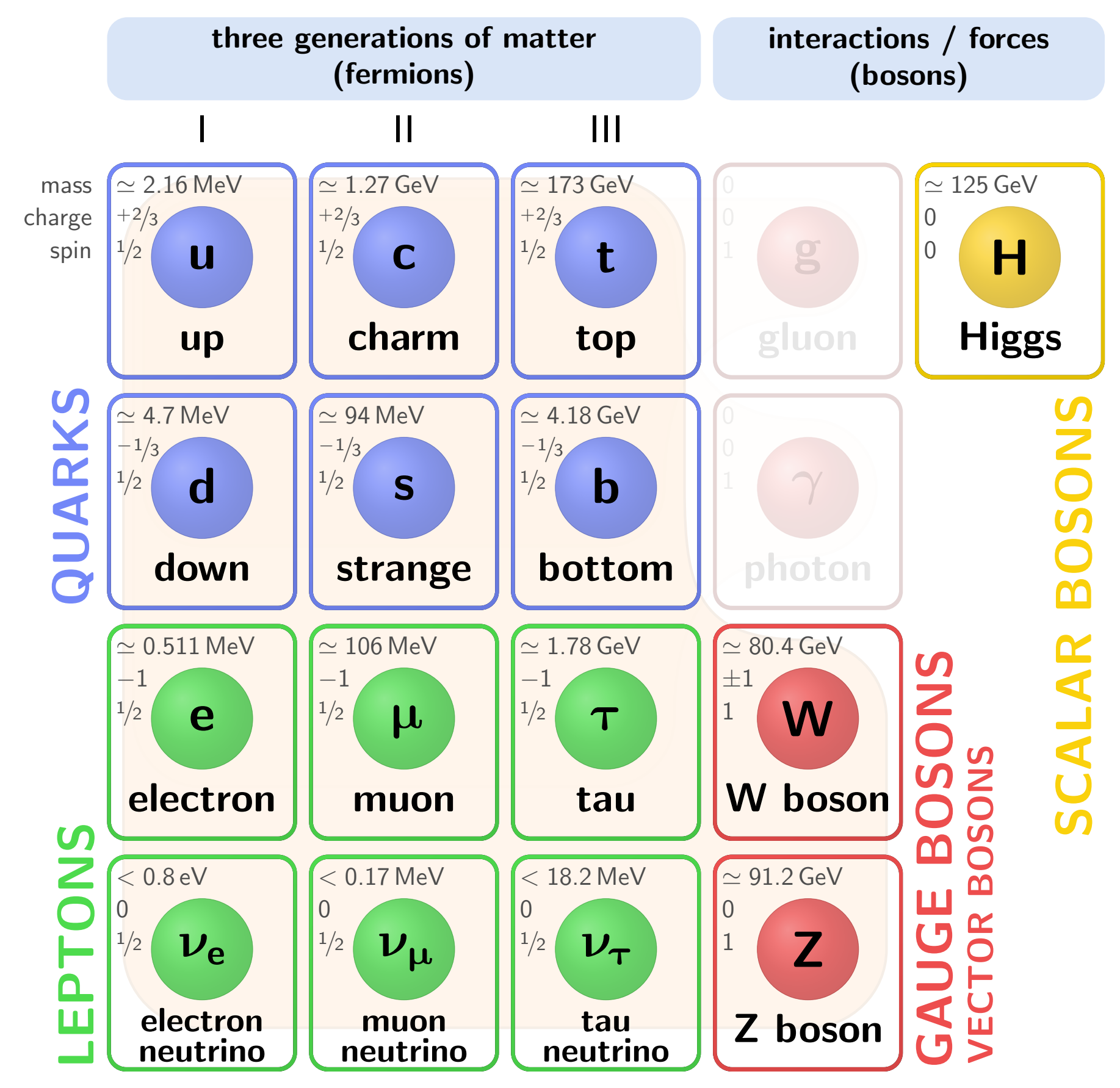

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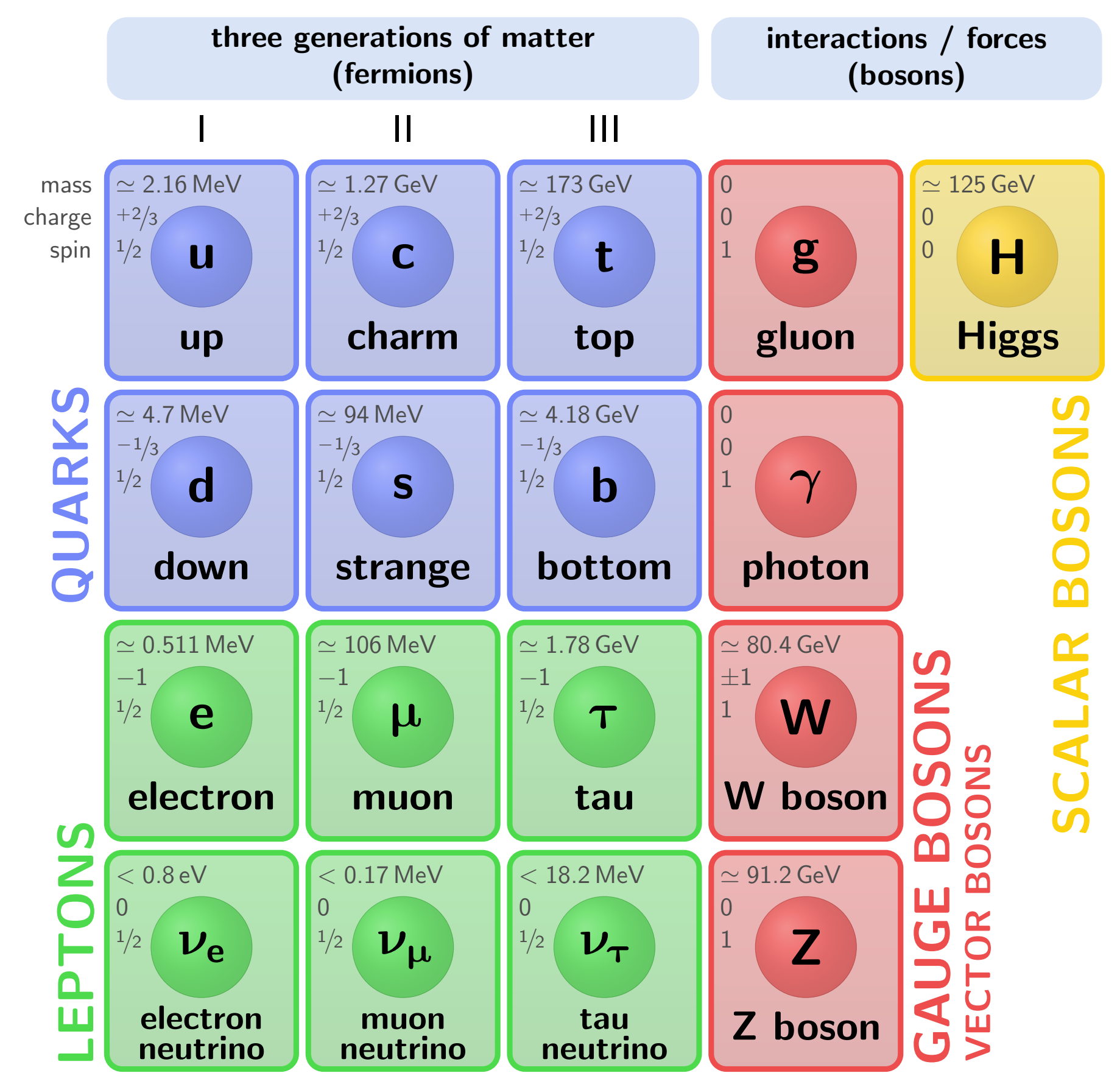

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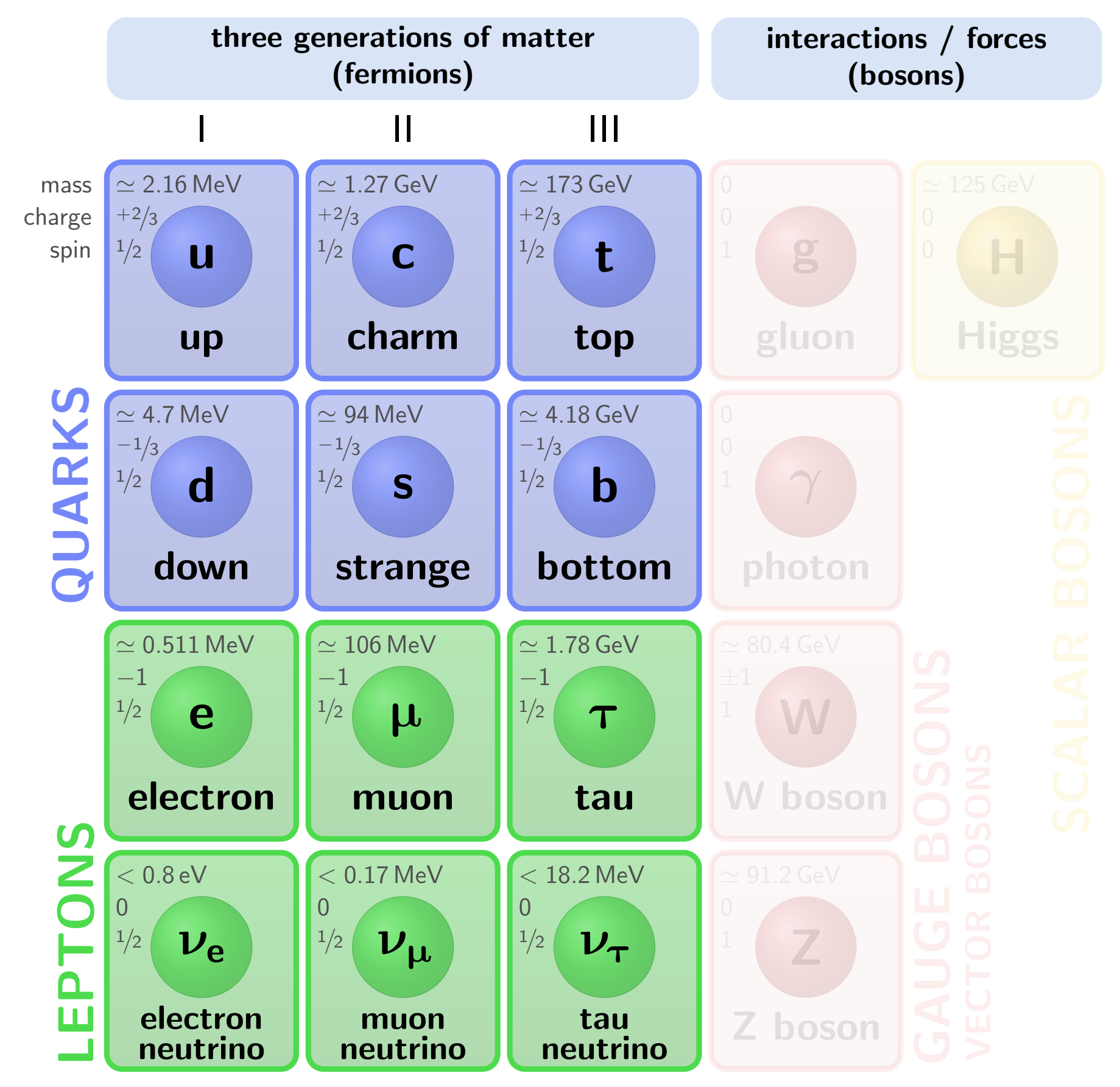

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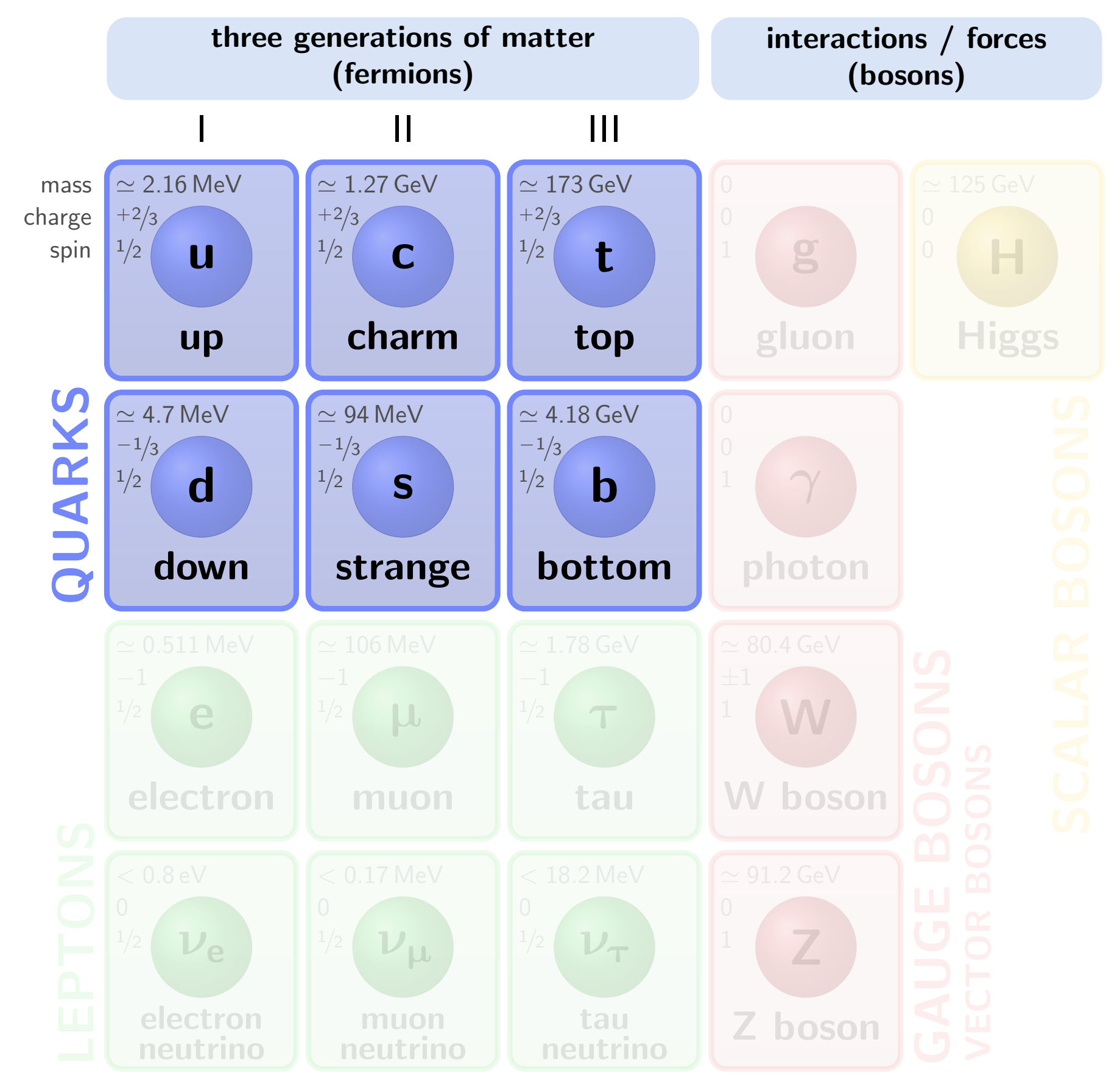

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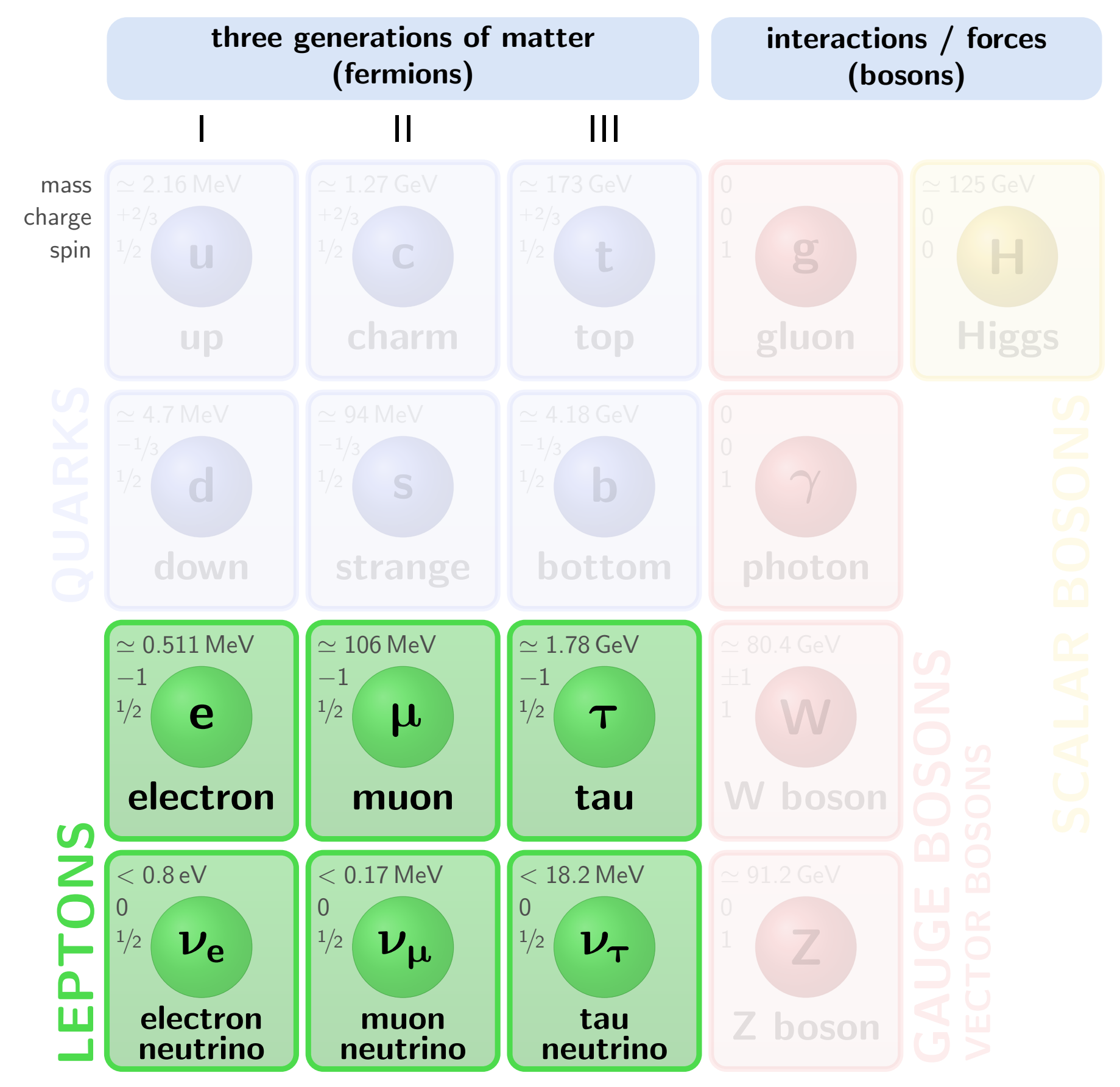

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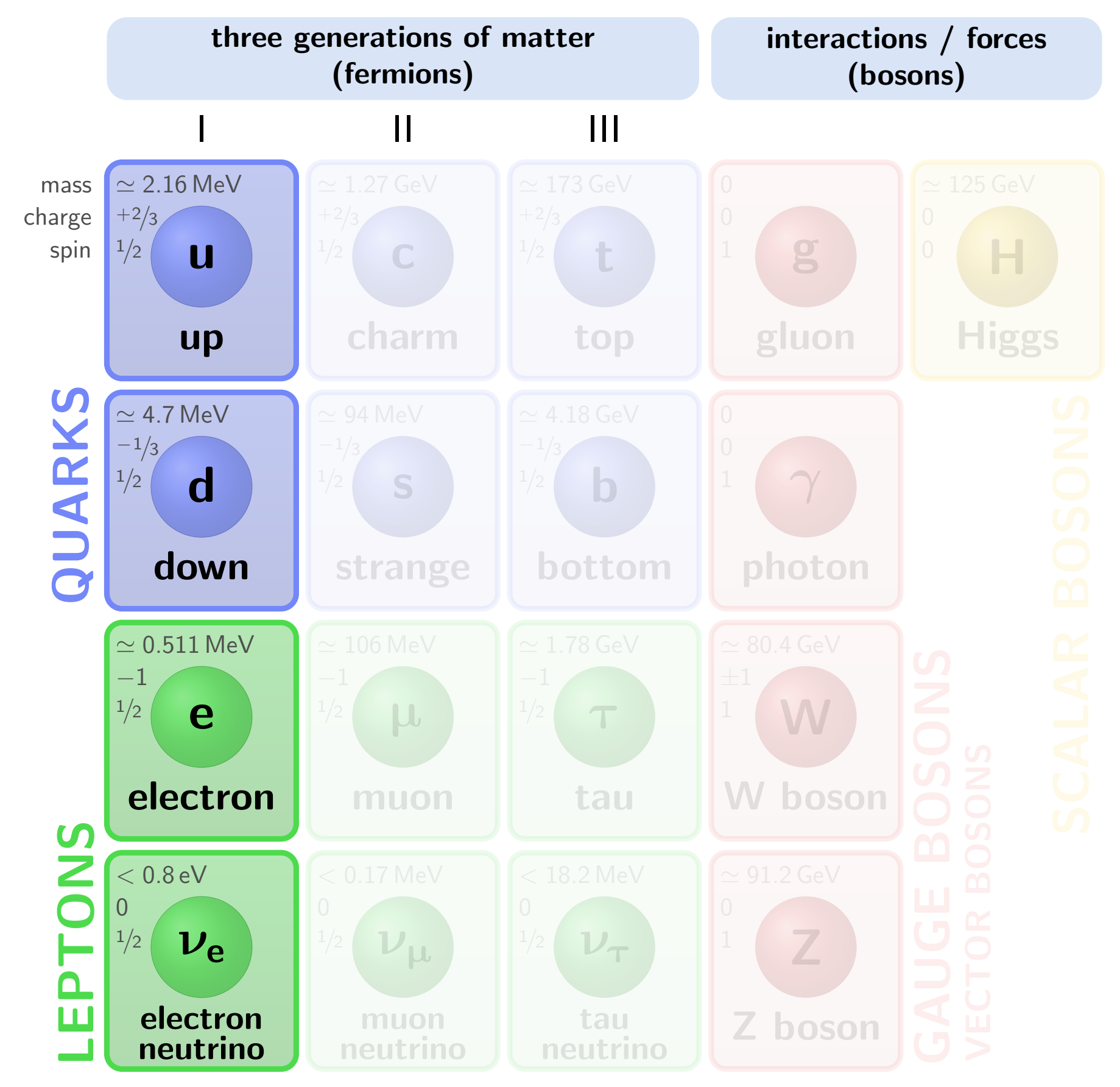

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0
0
1
γ
photon
SCALAR BOSONS
≃ 0.511 MeV
−1
1/2
e
electron
≃ 106 MeV
−1
1/2
μ
muon
≃ 1.78 GeV
−1
1/2
τ
tau
≃ 80.4 GeV
±1
1
W
W boson
GAUGE BOSONS
VECTOR BOSONS
LEPTONS
< 0.8 eV
0
1/2
νe
electron
neutrino
< 0.17 MeV
0
1/2
νμ
muon
neutrino
< 18.2 MeV
0
1/2
ντ
tau
neutrino
≃ 91.2 GeV
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Z
Z boson

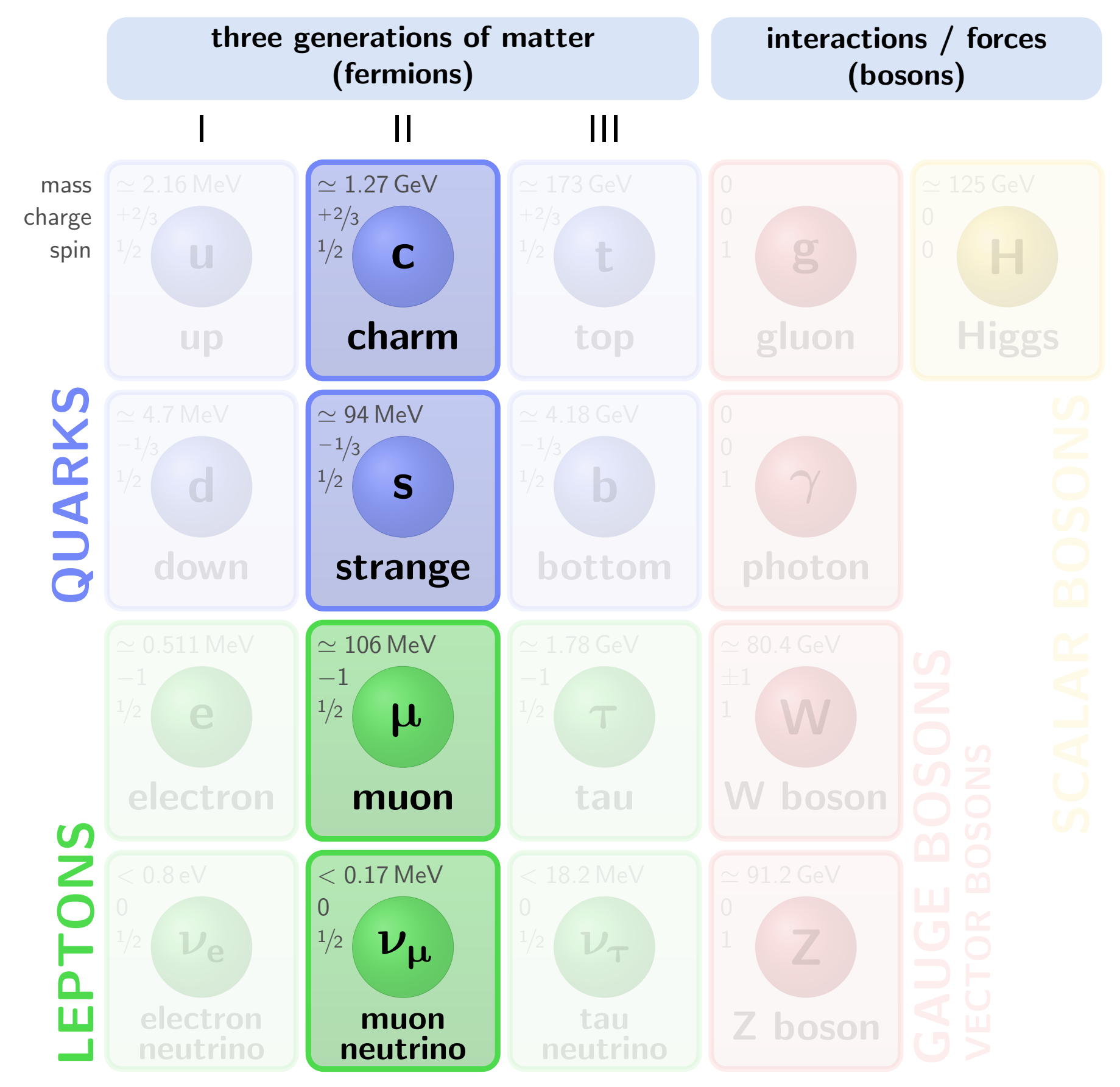

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
0
0
1
g
gluon
≃ 125 GeV
0
0
H
Higgs
QUARKS
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
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1/2
b
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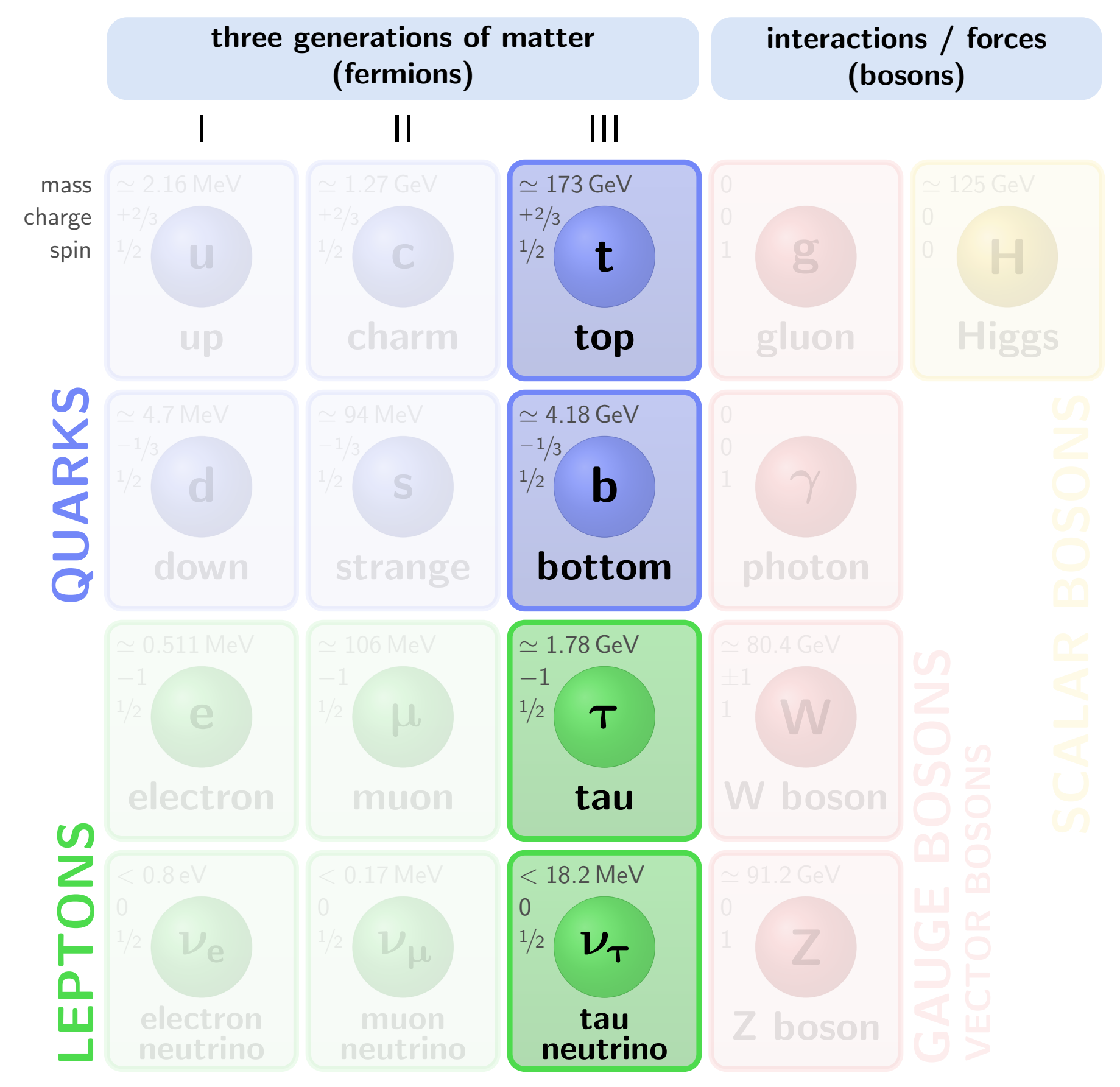

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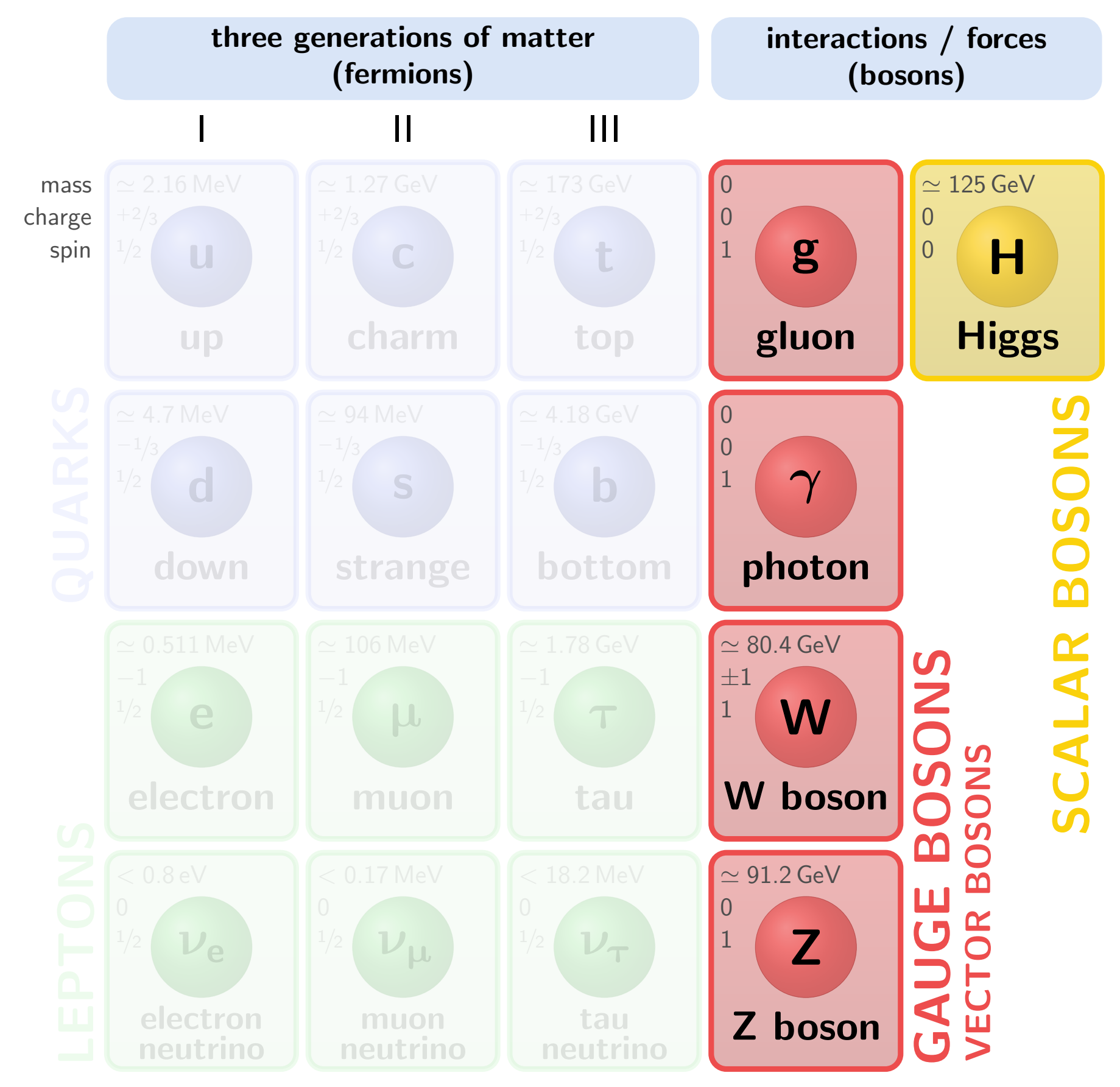

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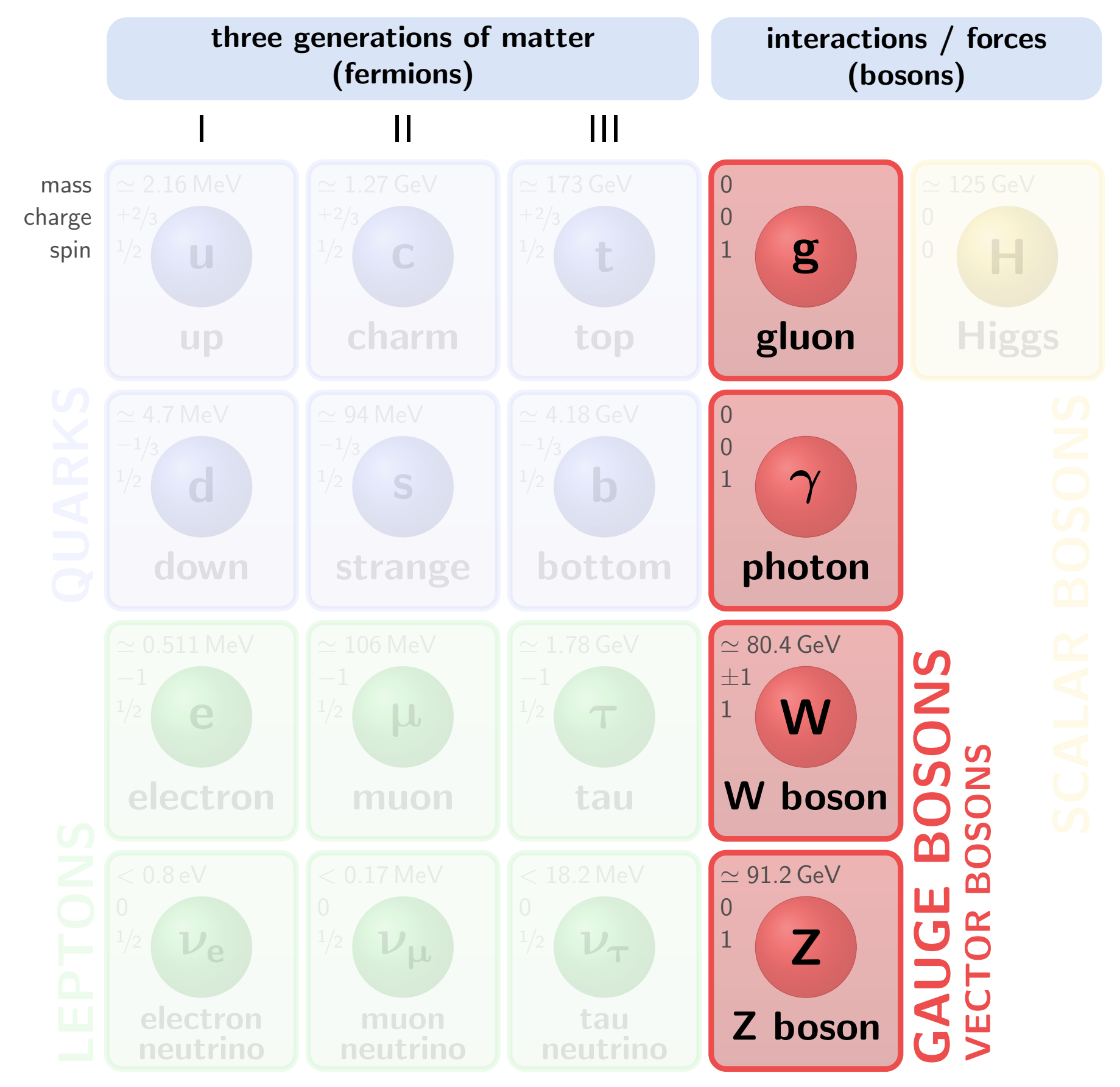

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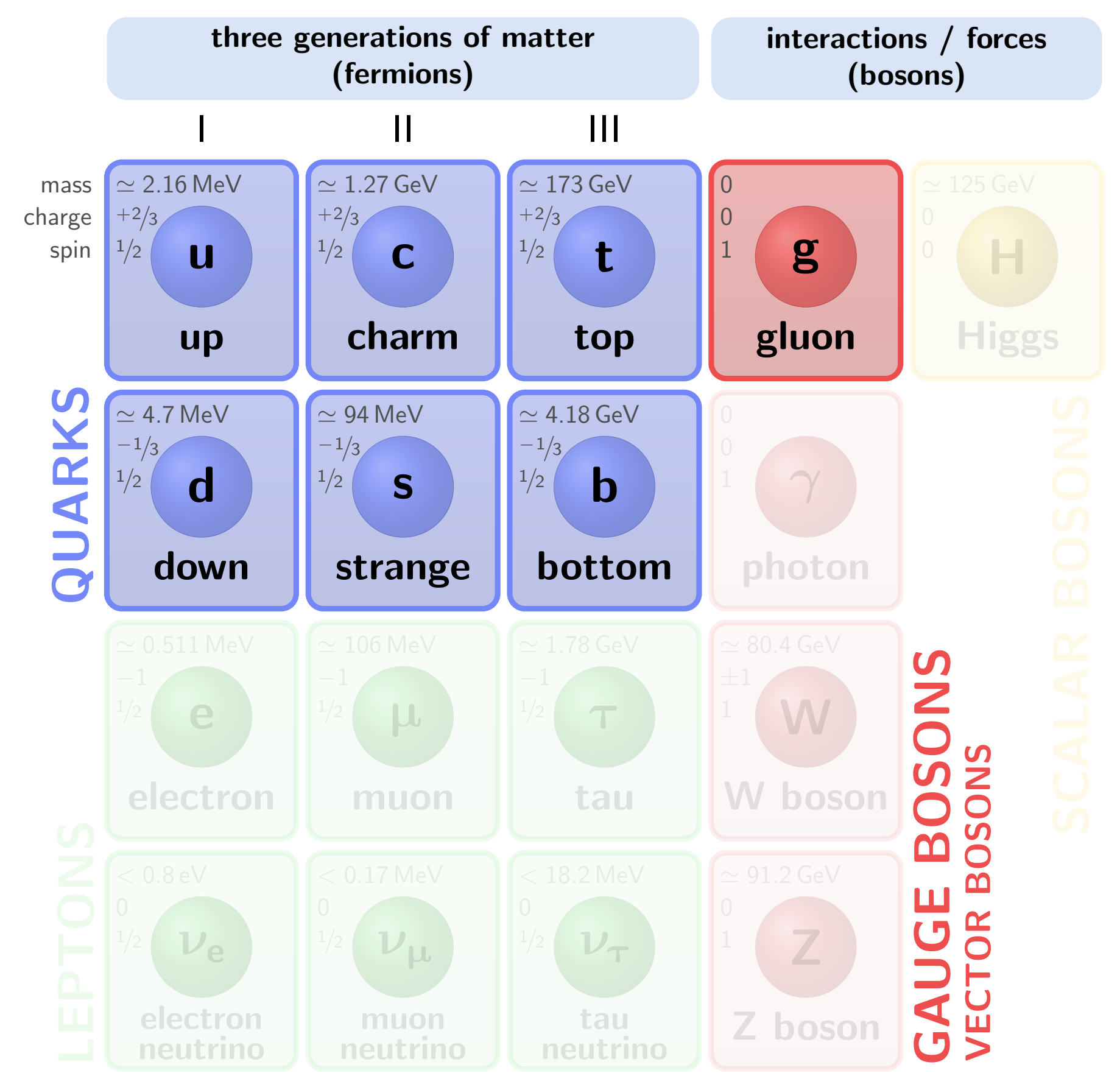

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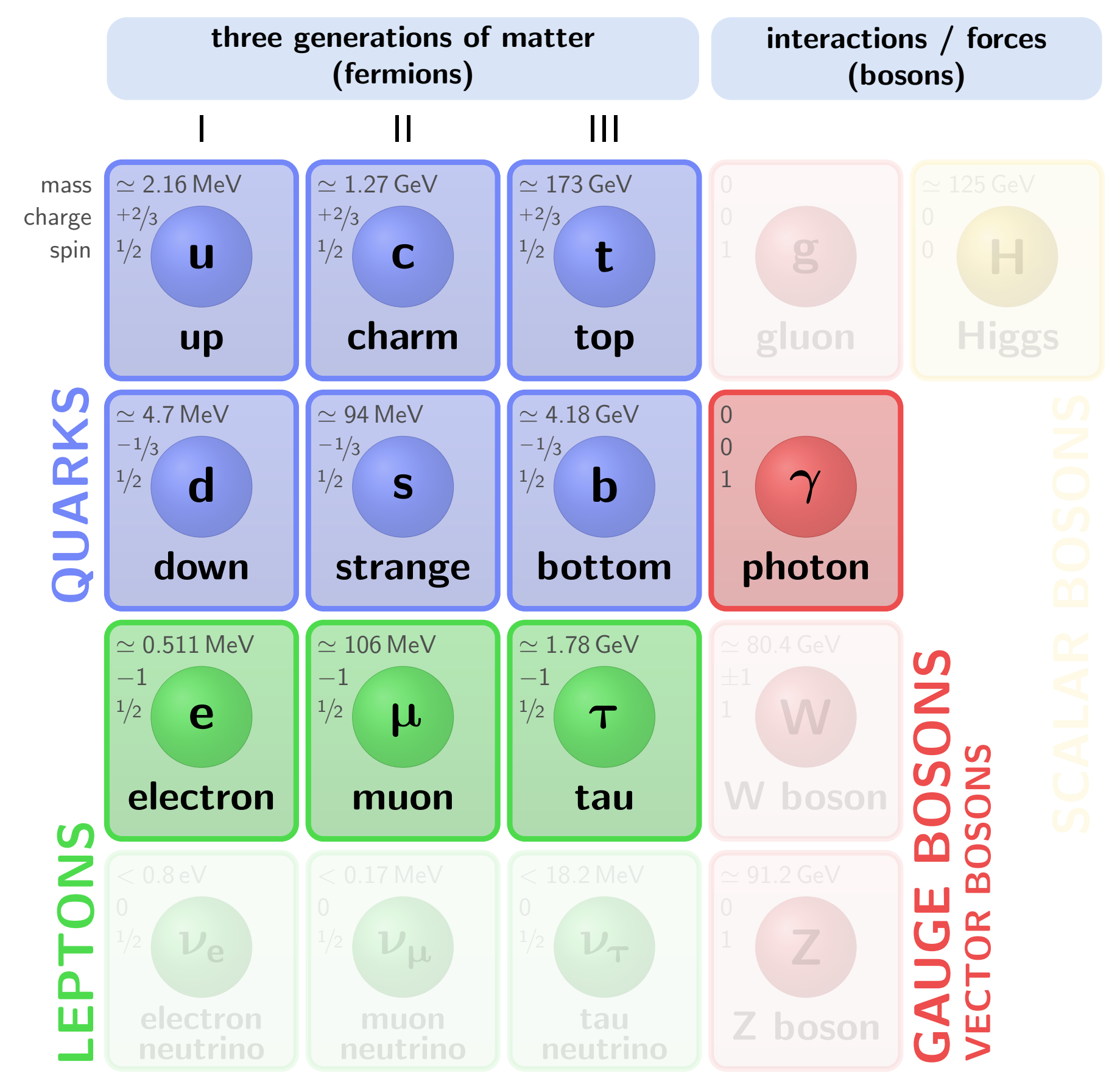

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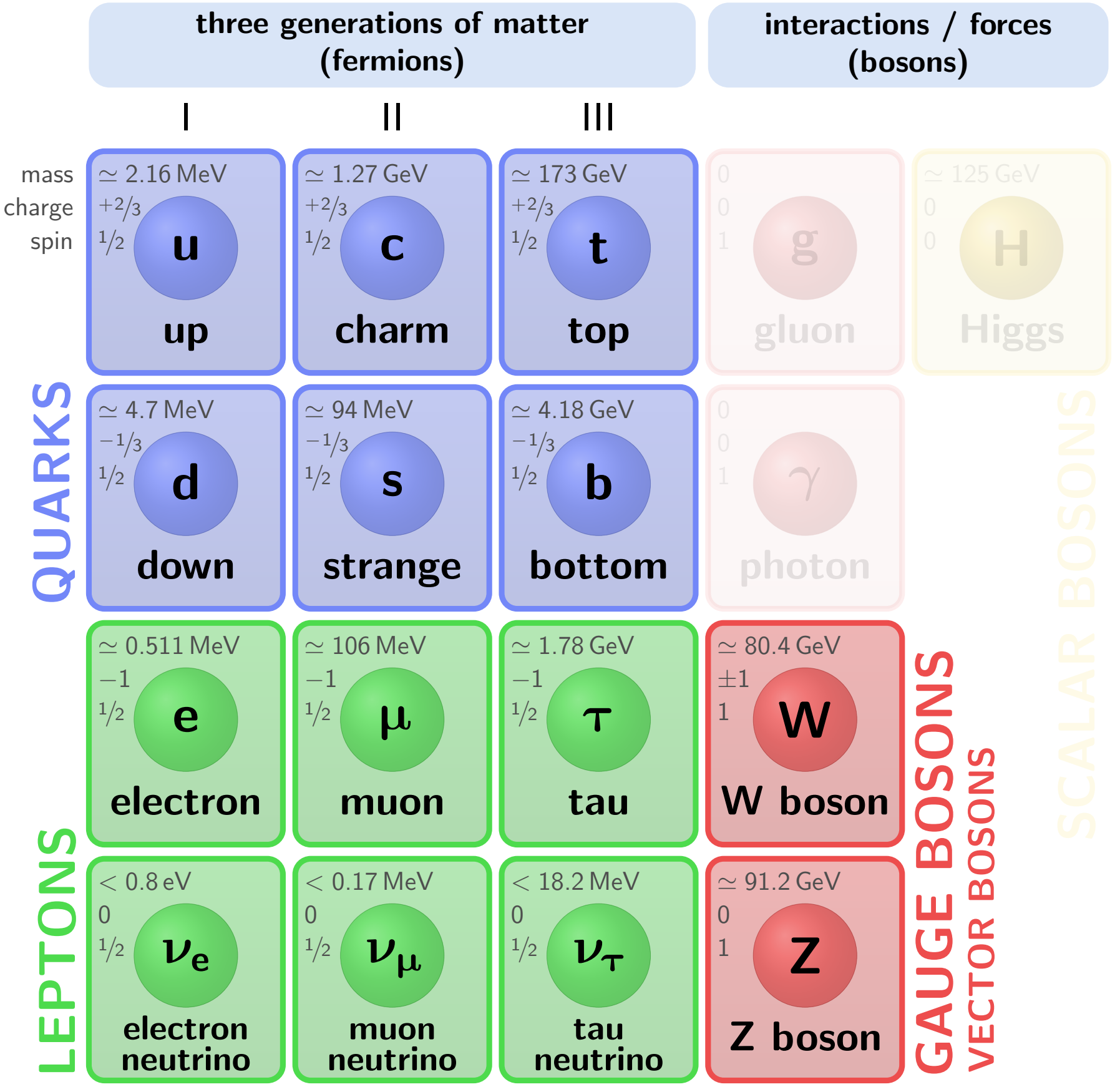

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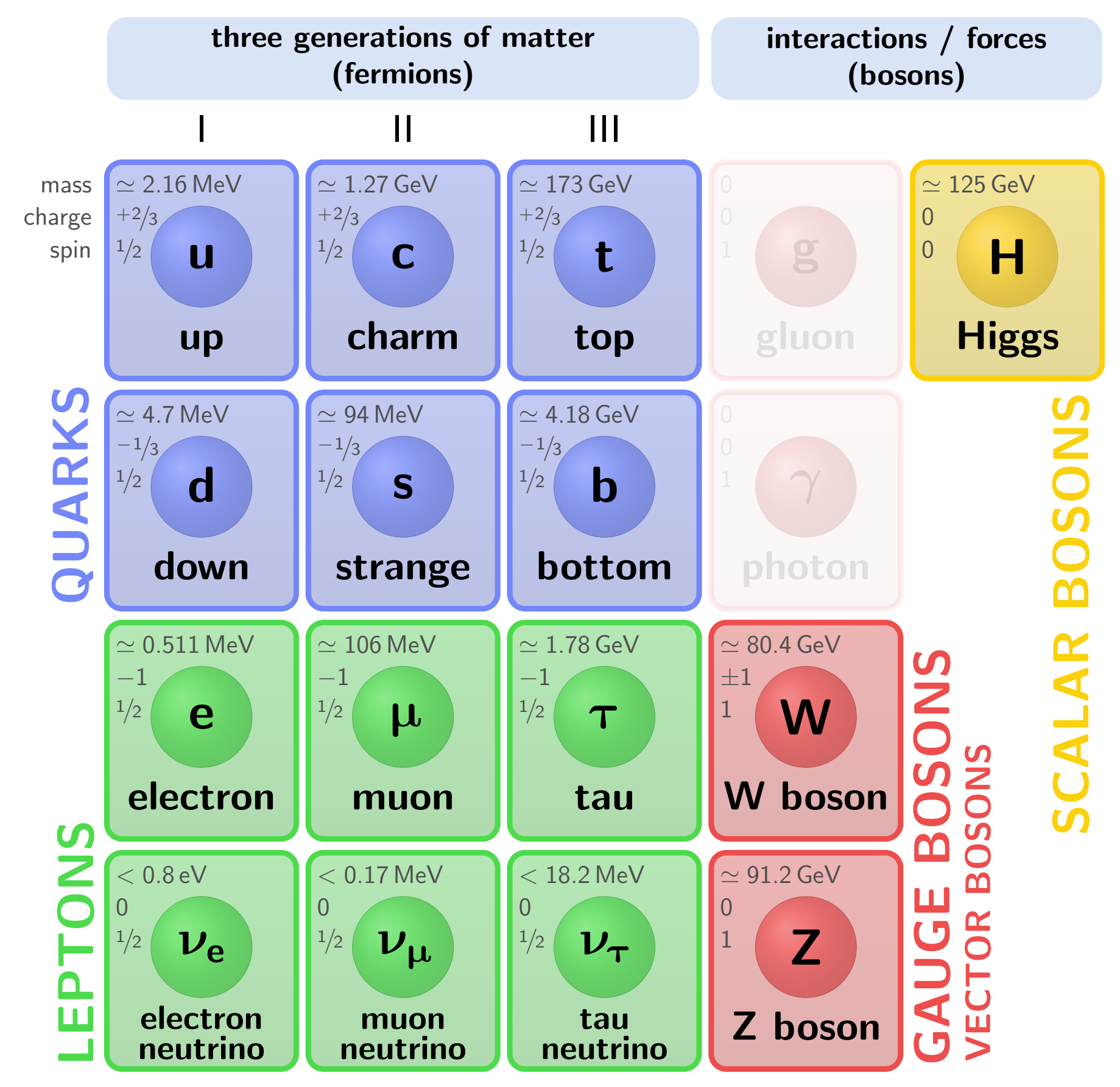

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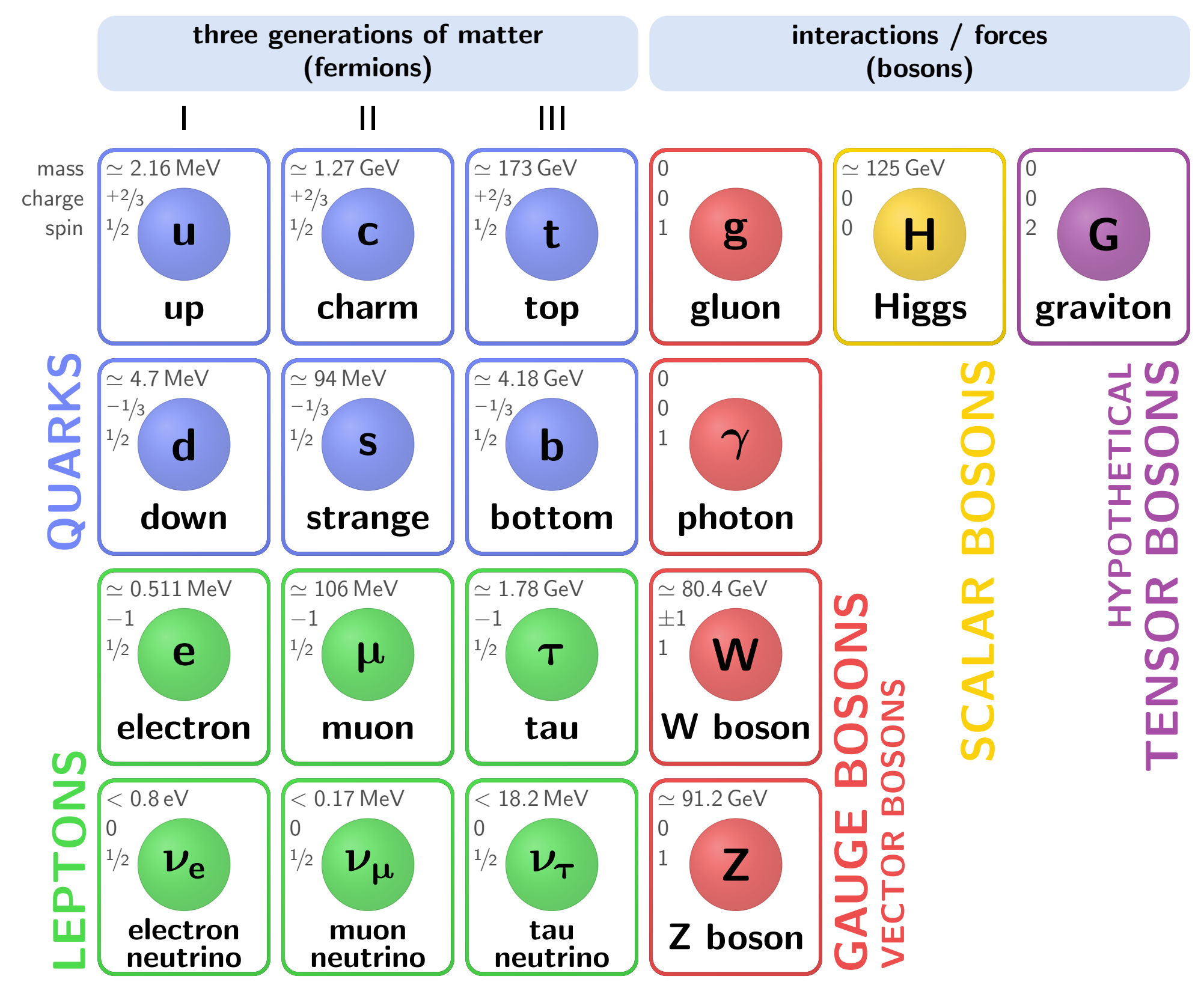

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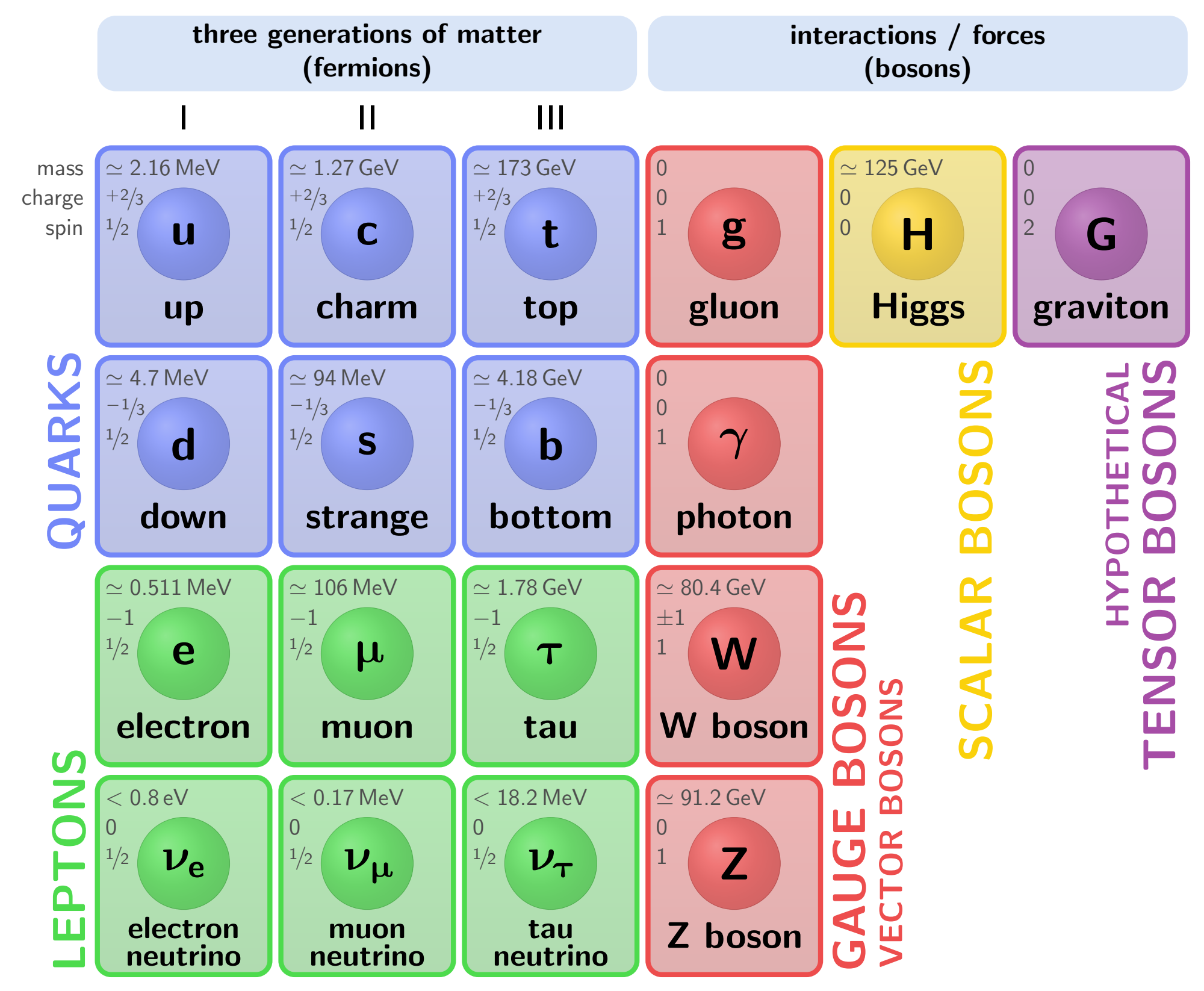

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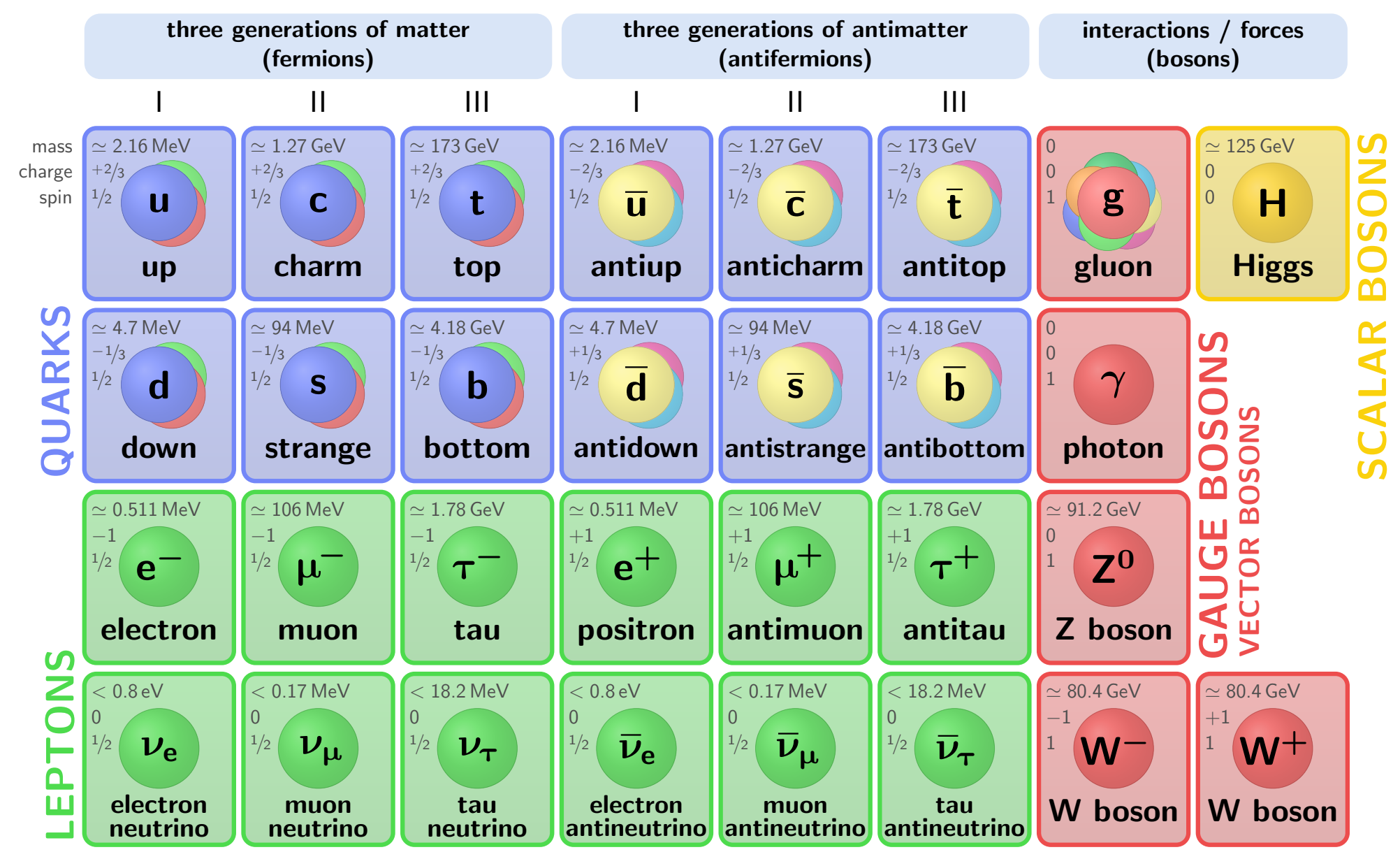

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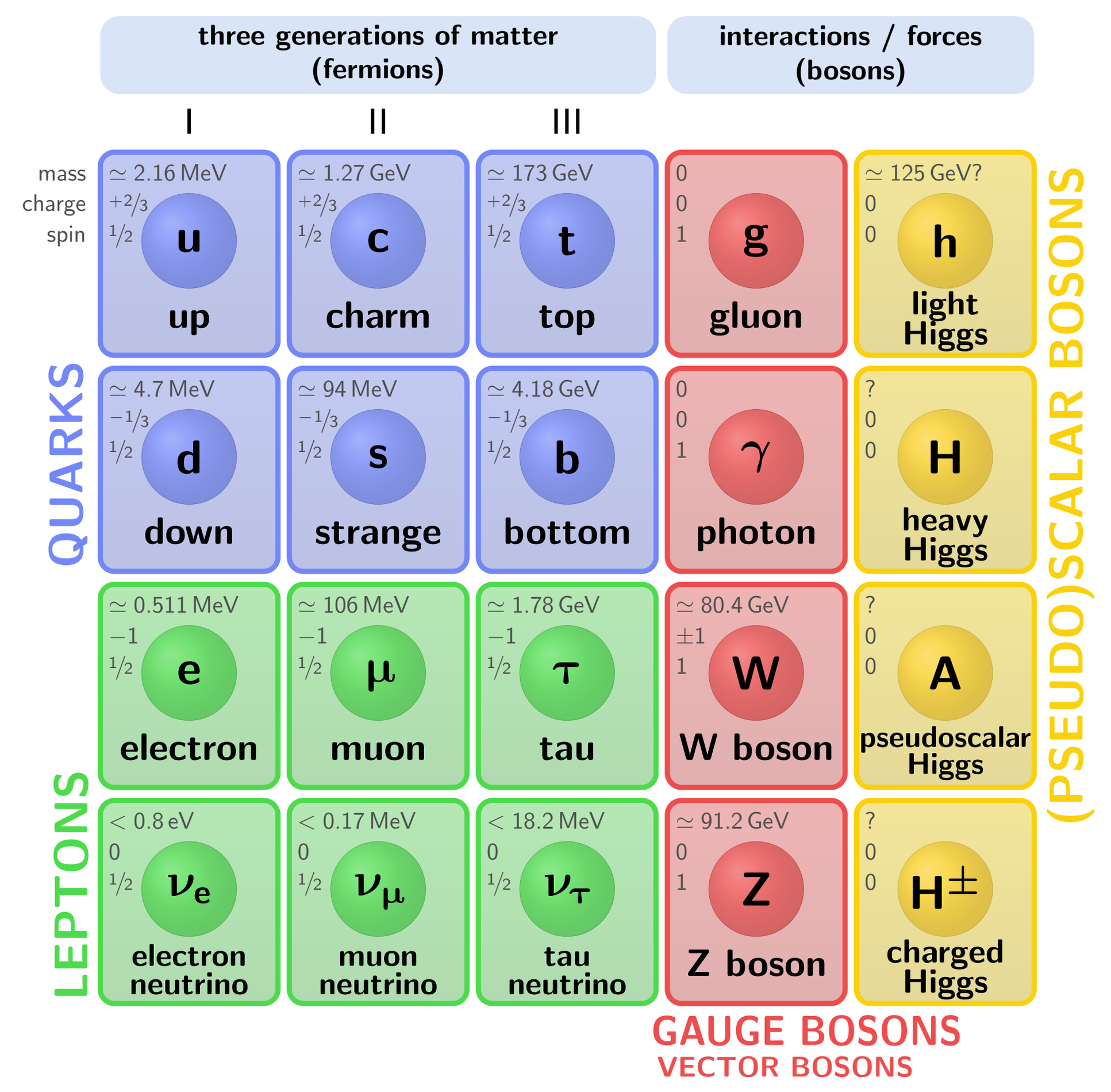

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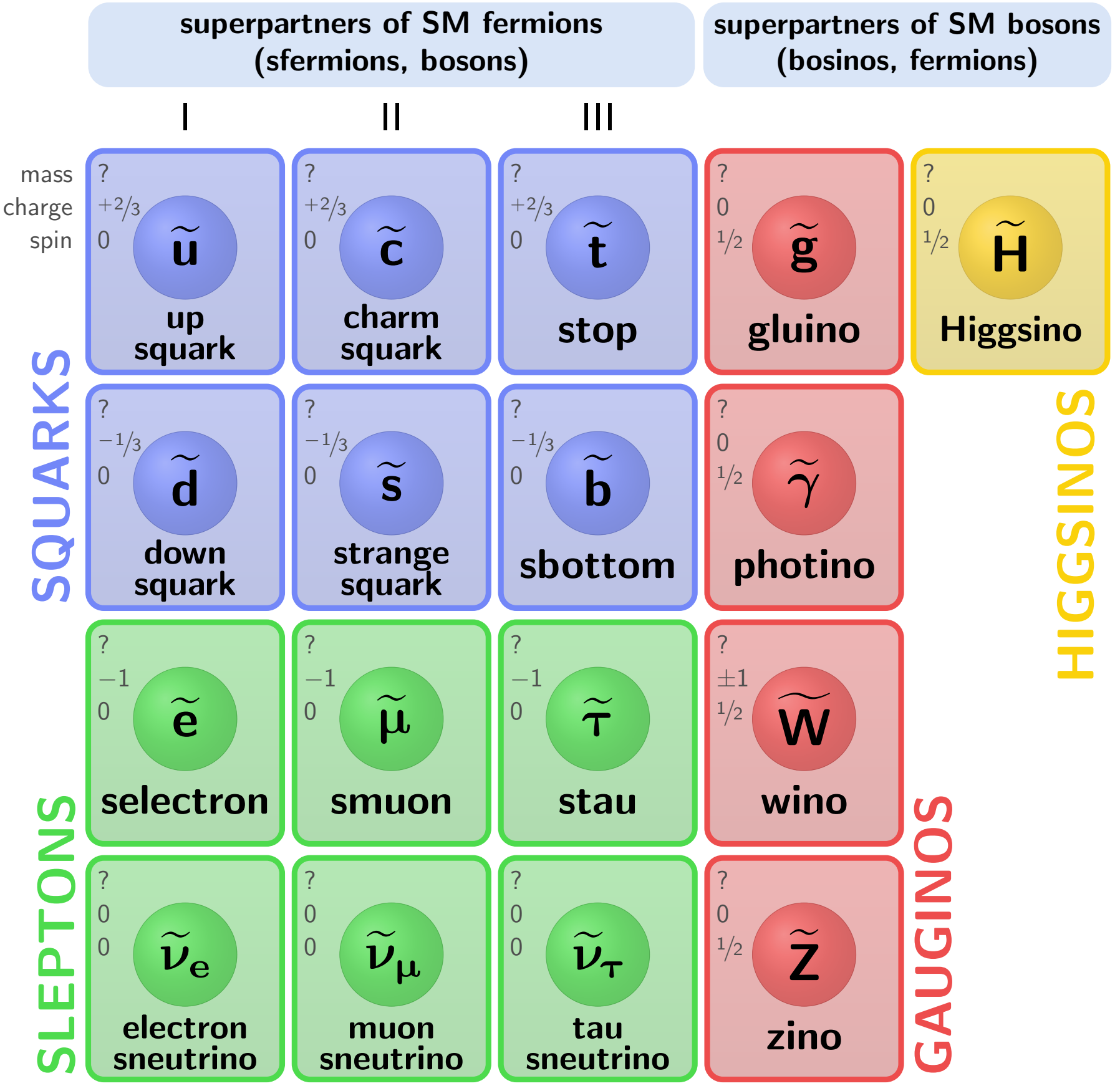

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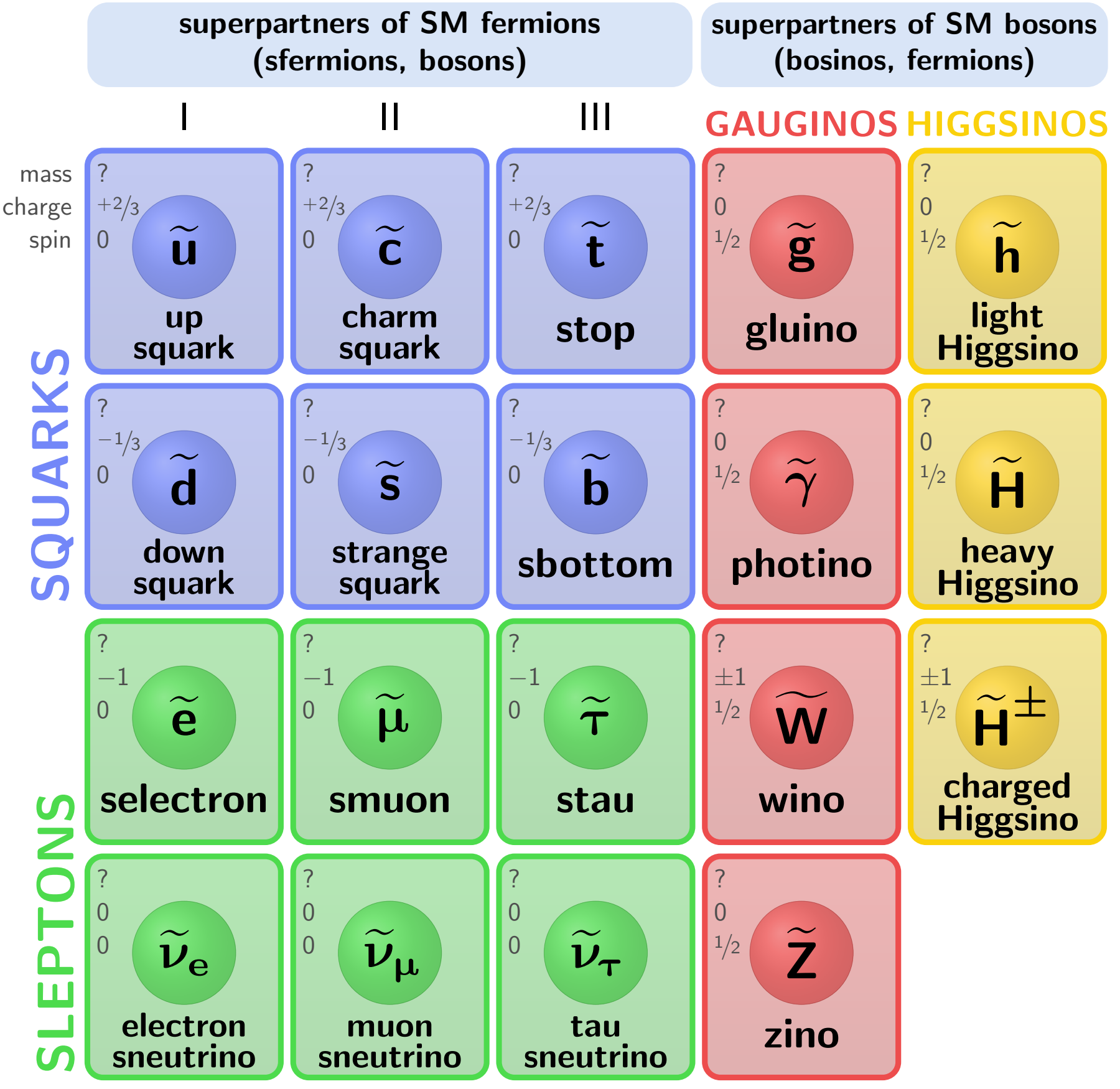

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I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
0
0
1
g
gluon
≃ 125 GeV
0
0
H
Higgs
0
0
2
G
graviton
QUARKS
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
−1/3
1/2
b
bottom
0
0
1
γ
photon
SCALAR BOSONS
HYPOTHETICAL
TENSOR BOSONS
≃ 0.511 MeV
−1
1/2
e
electron
≃ 106 MeV
−1
1/2
μ
muon
≃ 1.78 GeV
−1
1/2
τ
tau
≃ 80.4 GeV
±1
1
W
W boson
GAUGE BOSONS
VECTOR BOSONS
LEPTONS
< 0.8 eV
0
1/2
νe
electron
neutrino
< 0.17 MeV
0
1/2
νμ
muon
neutrino
< 18.2 MeV
0
1/2
ντ
tau
neutrino
≃ 91.2 GeV
0
1
Z
Z boson

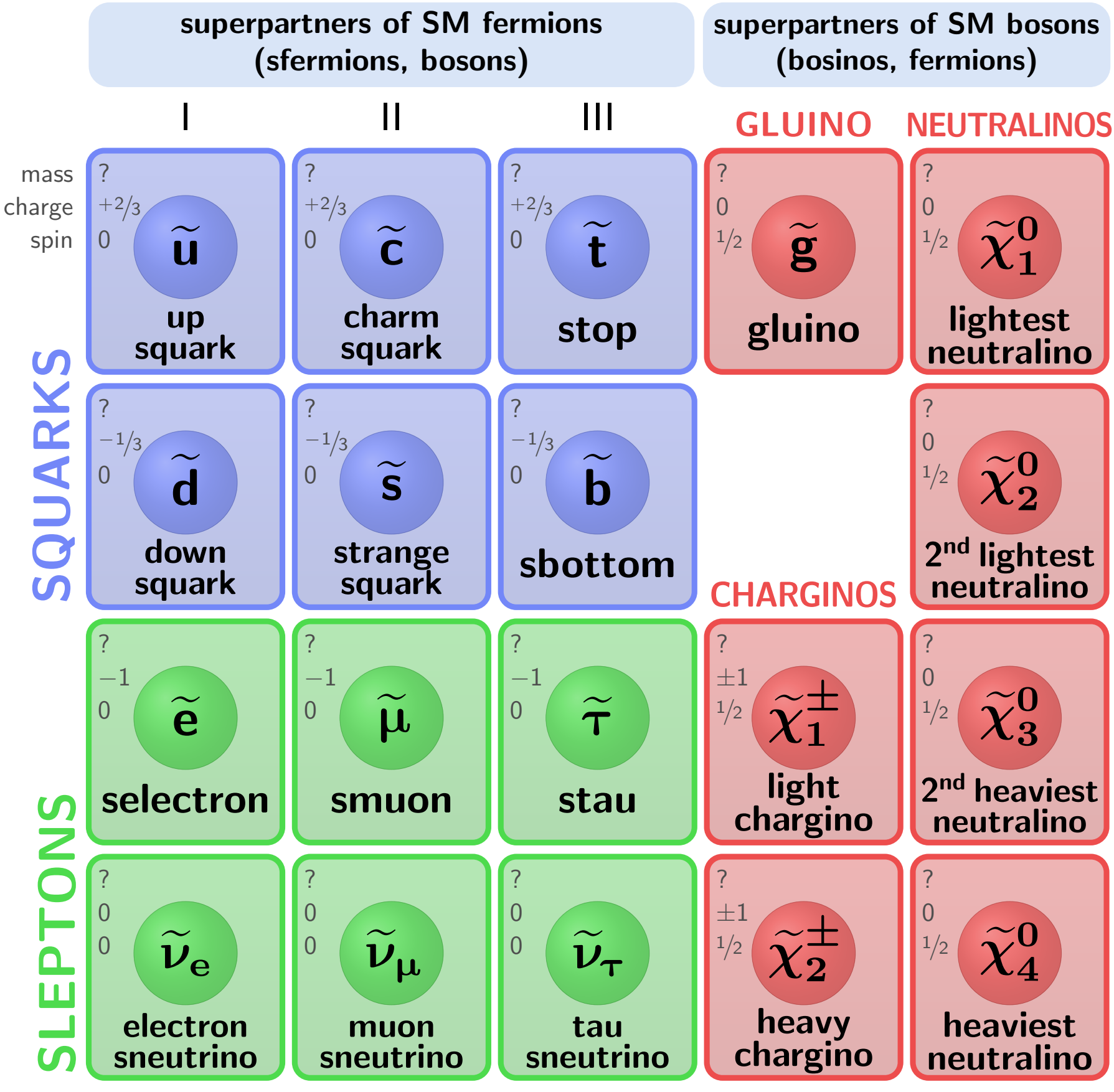

three generations of matter
(fermions)
interactions / forces
(bosons)
I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
0
0
1
g
gluon
≃ 125 GeV
0
0
H
Higgs
0
0
2
G
graviton
QUARKS
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
−1/3
1/2
b
bottom
0
0
1
γ
photon
SCALAR BOSONS
HYPOTHETICAL
TENSOR BOSONS
≃ 0.511 MeV
−1
1/2
e
electron
≃ 106 MeV
−1
1/2
μ
muon
≃ 1.78 GeV
−1
1/2
τ
tau
≃ 80.4 GeV
±1
1
W
W boson
GAUGE BOSONS
VECTOR BOSONS
LEPTONS
< 0.8 eV
0
1/2
νe
electron
neutrino
< 0.17 MeV
0
1/2
νμ
muon
neutrino
< 18.2 MeV
0
1/2
ντ
tau
neutrino
≃ 91.2 GeV
0
1
Z
Z boson


three generations of matter (fermions)
three generations of antimatter (antifermions)
interactions / forces (bosons)
I
II
III
I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
≃ 2.16 MeV
−2/3
1/2
ū
antiup
≃ 1.27 GeV
−2/3
1/2
c̄
anticharm
≃ 173 GeV
−2/3
1/2
t̄
antitop
0
0
1
g
gluon
≃ 125 GeV
0
0
H
Higgs
SCALAR BOSONS
QUARKS
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
−1/3
1/2
b
bottom
≃ 4.7 MeV
+1/3
1/2
d̄
antidown
≃ 94 MeV
+1/3
1/2
s̄
antistrange
≃ 4.18 GeV
+1/3
1/2
b̄
antibottom
0
0
1
γ
photon
GAUGE BOSONS
VECTOR BOSONS
≃ 0.511 MeV
−1
1/2
e−
electron
≃ 106 MeV
−1
1/2
μ−
muon
≃ 1.78 GeV
−1
1/2
τ−
tau
≃ 0.511 MeV
+1
1/2
e+
positron
≃ 106 MeV
+1
1/2
μ+
antimuon
≃ 1.78 GeV
+1
1/2
τ+
antitau
≃ 91.2 GeV
0
1
Z0
Z boson
LEPTONS
< 0.8 eV
0
1/2
νe
electron neutrino
< 0.17 MeV
0
1/2
νμ
muon neutrino
< 18.2 MeV
0
1/2
ντ
tau neutrino
< 0.8 eV
0
1/2
ν̄e
electron antineutrino
< 0.17 MeV
0
1/2
ν̄μ
muon antineutrino
< 18.2 MeV
0
1/2
ν̄τ
tau antineutrino
≃ 80.4 GeV
−1
1
W−
W boson
≃ 80.4 GeV
+1
1
W+
W boson


three generations of matter (fermions)
interactions / forces (bosons)
I
II
III
mass
charge
spin
≃ 2.16 MeV
+2/3
1/2
u
up
≃ 1.27 GeV
+2/3
1/2
c
charm
≃ 173 GeV
+2/3
1/2
t
top
0
0
1
g
gluon
≃ 125 GeV?
0
0
h
light Higgs
≃ 4.7 MeV
−1/3
1/2
d
down
≃ 94 MeV
−1/3
1/2
s
strange
≃ 4.18 GeV
−1/3
1/2
b
bottom
0
0
1
γ
photon
?
0
0
H
heavy Higgs
QUARKS
≃ 0.511 MeV
−1
1/2
e
electron
≃ 106 MeV
−1
1/2
μ
muon
≃ 1.78 GeV
−1
1/2
τ
tau
≃ 80.4 GeV
±1
1
W
W boson
?
0
0
A
pseudoscalar Higgs
(PSEUDO)SCALAR BOSONS
LEPTONS
< 0.8 eV
0
1/2
νe
electron neutrino
< 0.17 MeV
0
1/2
νμ
muon neutrino
< 18.2 MeV
0
1/2
ντ
tau neutrino
≃ 91.2 GeV
0
1
Z
Z boson
?
0
0
H±
charged Higgs
GAUGE BOSONS
VECTOR BOSONS


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
mass
charge
spin
SQUARKS
SLEPTONS
GAUGINOS
HIGGSINOS
? +2/3 0 ũ up squark
? +2/3 0 c̃ charm squark
? +2/3 0 t̃ stop
? 0 1/2 g̃ gluino
? 0 1/2 H̃ Higgsino
? −1/3 0 d̃ down squark
? −1/3 0 s̃ strange squark
? −1/3 0 b̃ sbottom
? 0 1/2 γ̃ photino
? −1 0 ẽ selectron
? −1 0 μ̃ smuon
? −1 0 τ̃ stau
? ±1 1/2 W̃ wino
? 0 0 ν̃e electron sneutrino
? 0 0 ν̃μ muon sneutrino
? 0 0 ν̃τ tau sneutrino
? 0 1/2 Z̃ zino


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
GAUGINOS
HIGGSINOS
mass
charge
spin
SQUARKS
SLEPTONS
? +2/3 0 ũ up squark
? +2/3 0 c̃ charm squark
? +2/3 0 t̃ stop
? 0 1/2 g̃ gluino
? 0 1/2 h̃ light Higgsino
? −1/3 0 d̃ down squark
? −1/3 0 s̃ strange squark
? −1/3 0 b̃ sbottom
? 0 1/2 γ̃ photino
? 0 1/2 H̃ heavy Higgsino
? −1 0 ẽ selectron
? −1 0 μ̃ smuon
? −1 0 τ̃ stau
? ±1 1/2 W̃ wino
? ±1 1/2 H̃± charged Higgsino
? 0 0 ν̃e electron sneutrino
? 0 0 ν̃μ muon sneutrino
? 0 0 ν̃τ tau sneutrino
? 0 1/2 Z̃ zino


superpartners of SM fermions
(sfermions, bosons)
superpartners of SM bosons
(bosinos, fermions)
I
II
III
GLUINO
NEUTRALINOS
mass
charge
spin
SQUARKS
SLEPTONS
CHARGINOS
? +2/3 0 ũ up squark
? +2/3 0 c̃ charm squark
? +2/3 0 t̃ stop
? 0 1/2 g̃ gluino
? 0 1/2 χ̃₁⁰ lightest neutralino
? −1/3 0 d̃ down squark
? −1/3 0 s̃ strange squark
? −1/3 0 b̃ sbottom
? 0 1/2 χ̃₂⁰ 2nd lightest neutralino
? −1 0 ẽ selectron
? −1 0 μ̃ smuon
? −1 0 τ̃ stau
? ±1 1/2 χ̃₁± light chargino
? 0 1/2 χ̃₃⁰ 2nd heaviest neutralino
? 0 0 ν̃e electron sneutrino
? 0 0 ν̃μ muon sneutrino
? 0 0 ν̃τ tau sneutrino
? ±1 1/2 χ̃₂± heavy chargino
? 0 1/2 χ̃₄⁰ heaviest neutralino